

How Artificial Intelligence Shapes Science: Evidence from AlphaFold *

Ryan Hill, Northwestern University
Carolyn Stein, UC Berkeley

July 17, 2026

Abstract

We study how a frontier AI model affects scientific discovery by examining the release of the AlphaFold2 algorithm and its impact on structural biology and related fields of science. Structural biology is the field of science concerned with understanding the structure of proteins. Researchers in this field historically devoted substantial time and resources to experimentally solving three-dimensional protein structures. AlphaFold can *predict* these structures without running experiments. In July 2021, researchers gained access to hundreds of thousands of these AI-predicted structures virtually overnight. Yet, to date, we find that the rate of experimental structure determination has remained almost unchanged. Instead, researchers appear to use predicted structures to facilitate and complement experimental structure determination. Looking at downstream science that builds on protein structures, we find that basic research on proteins that had no structural information prior to AlphaFold increases by 23 to 32% relative to proteins that already had a structure, with the effects increasing dramatically over time. This represents a significant shift in the direction of research toward less-studied proteins. By contrast, the response of applied R&D, as measured by patents referencing those proteins, has been far more muted. *JEL Codes:* O33, O31, J24, I23, L65.

*We are especially grateful to Paul D. Adams, Susan Tsutakawa, Joanna Slusky, Jason Wagoner, and Brian Shoichet for providing expertise and insight into their respective fields of science, and to Sylvain Poux and Elisabeth Gasteiger for sharing their expertise on UniProt. We thank Daron Acemoglu, David Autor, Pierre Azoulay, Eric Budish, Florian Ederer, Daniel Gross, Anders Humlum, Benjamin Jones, Juan Mateos-Garcia, Emi Nakamura, Jeremy Stein, and seminar participants at Berkeley, Northwestern, University of Kansas, Johns Hopkins, University of Chicago, and Harvard Business School for comments that substantially improved the paper. Thomas Barden, Karan Makkar, and Alexia Witthaus Viñé provided excellent research assistance. All errors and omissions are our own. Contact: ryan.hill@kellogg.northwestern.edu, carolyn_stein@berkeley.edu. First version: March 11, 2026.

1 Introduction

Artificial intelligence (AI) is rapidly reshaping industries in real time. A growing body of evidence suggests that AI is automating an increasingly broad set of economic tasks, ranging from routine work in call centers (Brynjolfsson et al., 2025) to creative tasks in photography and digital design (Goldberg and Lam, 2025; Zhou and Lee, 2024), as well as high-skill occupations such as software engineering and radiology (Cui et al., 2025; Agarwal et al., 2023). In many cases, breakthroughs are improving productivity in the market for goods and services. At the same time, AI is also beginning to influence the production of ideas, which has tantalizing implications for the possibility of accelerated economic growth (Aghion et al., 2019). The non-rivalrous and cumulative nature of ideas suggests that the automation of research tasks can have amplifying effects for the productivity of the broader economy (Romer, 1990; Jones, 1995; Weitzman, 1998; Jones, 2005). Practically, however, the R&D pipeline is full of bottlenecks. The potential complementarity between tasks in a long and uncertain research pipeline may act as a natural damper on the pace of progress, even when AI fully automates many steps along the chain (Jones, 2025). Despite many promising anecdotes about the changing pace of AI-enabled R&D, there has been scant empirical evidence about how AI is changing the research process in practice.

In this paper, we provide some of the first systematic evidence on how AI affects scientific discovery, using the introduction of protein structure prediction models as an important case study. Specifically, we study AlphaFold2, a Nobel-prize winning machine learning algorithm that rapidly increased the availability of predicted structures relevant for a broad range of basic and applied research fields related to proteins.¹ This was a massive technology shock to structural biology, the field of research that seeks to determine the three-dimensional structures of proteins.² These structural models help us understand how proteins function inside of cells and how they might be targeted in pharmaceutical therapies. Since the early 1970s, we have relied on slow and expensive experimental methods to generate the roughly 130,000 unique publicly available protein structures available today. At the same time, advances in DNA sequencing have made it inexpensive and routine to identify the genetic makeup of proteins, leading to databases containing hundreds of millions of protein sequences—of which less than 0.1 percent have known structures. This gap motivates the longstanding *protein folding problem*: can we predict a protein’s three-dimensional structure from its genetic code,

¹ There is an earlier version of the algorithm, released in 2018, known as AlphaFold. However, because AlphaFold2 represented the primary advance, throughout this paper we will use the phrase “AlphaFold” to refer to AlphaFold2, unless explicitly stated otherwise.

² In recent years, several economics of science papers have used this field of science as an empirical setting, including Hill and Stein (2025b), Hill and Stein (2025a), Kim (2026), and Zhuo (2025).

without running any experiments?

For decades, computational biologists and computer scientists made slow progress on improving the capabilities of computational models. A breakthrough occurred when Google DeepMind debuted the AlphaFold2 algorithm in late 2020. For the first time, an AI model achieved near-experimental levels of accuracy at a vastly reduced cost in terms of time and resources, to the surprise of many in the research community. Figure 1 shows examples of two predicted protein structures overlaid with their experimental counterparts.

Many expert and casual observers characterized this advance as effectively solving the protein folding problem and anticipated far-reaching implications for both basic scientific research and medical innovation. One of the scientists tasked with assessing the algorithm’s performance, evolutionary biologist Andrei Lupas, memorably said of AlphaFold2, “This will change medicine. It will change research. It will change bioengineering. It will change everything.” Structure prediction tools may open up new paths for scientific research about protein function, molecular mechanisms, and cellular processes. They may also reshape drug discovery and medical R&D by illuminating important disease targets. AlphaFold therefore provides a natural setting to study both the narrow and broader downstream impacts of artificial intelligence on scientific progress.

While AlphaFold is not the only major AI advance in science, it presents a particularly compelling case study for several reasons. First is its timing. AlphaFold2 was first revealed in late 2020, with predicted structures made widely available in mid-2021. Thus, it arrived early compared to other major AI breakthroughs,³ giving us more time to understand its impact. Second, the breakthrough was unexpected. While the first version of AlphaFold, launched in 2018, outperformed competitors, it was not considered good enough to replace experimental structure determination. The breakthrough improvement in AlphaFold2 was unexpected by the scientific community, and led experts to declare that the protein folding problem had been solved. Conference attendees who saw the initial reports of AlphaFold2’s prediction accuracy claimed to be “in shock” (Ball, 2024). Third, the nature of the algorithm is particularly clean and concrete. Compared to other AI tools—such as large-language models, coding agents, etc.—it is unusually clear what tasks AlphaFold is potentially automating. This makes it easier to interpret subsequent shifts in research pace and direction. Finally, it is an important application. Structural biology and protein folding is a critical area of science. Several Nobel Prizes have been awarded for the discovery of a *single* experimental protein structure, and these structures enable important advances in the understanding of biological processes, disease, and drug design. Moreover, the AlphaFold2 algorithm was awarded the 2024 Nobel Prize, an unusually early indicator of its perceived scientific importance.

³ For example, the first iteration of ChatGPT did not launch until November 2022.

The goal of this paper is to understand how this shock impacted the production of science, both in the narrow field of experimental structural biology, and in downstream fields of science and R&D that build on these structures. We first present evidence on how the introduction of AlphaFold affected experimental structural biology. For researchers in this field, the arrival of AlphaFold had the potential to dramatically reshape the production technology for structure determination. Yet we find limited evidence that it has substituted for experimental structure determination thus far. Using the universe of experimental protein structures released in the Protein Data Bank (PDB), we find that since the introduction of AlphaFold and the first wave of predicted structures in 2021, there has been a slight *increase* in the number of experimental structures being solved and uploaded to the PDB. Moreover, the number of papers reporting structural biology experiments has also stayed consistent, including in papers that are published in the top general-interest science journals.

By combining novel survey data from active structural biologists with data from the PDB, we propose several reasons for why scientists continue to solve experimental structures. First, structural biologists report that predicted structures can be unreliable, and that the vast majority say they do not trust predicted structures enough to skip experimental validation. Thus, it may be the case that low (or unpredictable) prediction quality continues to make experimental structures valuable. Second, AlphaFold still struggles to predict certain types of proteins. Around half of the researchers in our survey sample report that they pivot to working on protein structures that AlphaFold cannot predict well, suggesting a more narrow form of substitution. We find some supporting evidence of this pivoting using data from the PDB. We caveat that researcher incentives may also play a role: Expert scientists, many of whom are tenured, may prefer to continue doing the science they are uniquely trained to perform. Still, our surveyed scientists report that they expect to solve even more experimental structures by 2031, suggesting that they continue to view experiments as valuable, even as prediction models are expected to improve.

Instead of substitution, we find strong evidence that structural biologists are using AlphaFold to *complement* their experimental work. Experimental structure prediction involves computational steps that can be accomplished faster and more accurately using existing structures as a template. With these methods, researchers can use the template as a starting point in their three-dimensional model, rather than building it entirely from scratch. In the past, scientists would rely on similar experimental structures as templates, limiting their ability to use these more efficient methods in exploratory work about novel proteins. However, AI-predicted structures can also be used as templates. After AlphaFold, we see a sharp uptick in the use of these template-based methods, concentrated among structures that lack a similar experimental structure. This complementarity between experimental and

AI-based research may open opportunities for greater exploration of the protein space.

Second, we turn our attention to the broader impacts of AlphaFold on related disciplines and downstream pharmaceutical R&D. Here we exploit the insight that only a small share of known proteins had an experimental structure model prior to AlphaFold. This creates a natural experiment at the protein level. Proteins with an experimentally solved structure prior to AlphaFold did not experience much—if any—new structural information from the arrival of predicted structures. On the other hand, for proteins *without* an experimentally solved structure, AlphaFold represented a large shock of new structural information. Thus, we can conceptualize the previously solved proteins as the “control” group and the previously unsolved proteins as the “treated” group. Then we can compare protein-level research activity in broader and downstream fields across these two groups in a difference-in-differences design.

To do this, we use data from the Universal Protein Resource (UniProt), a database of all known proteins. Because of the scale (UniProt contains information on over 200 million proteins) we focus on the more curated 570,000 protein subsample of this data source, known as SwissProt. This includes proteins that are of higher scientific importance (and includes all human proteins). By combining these data with the PDB, we are able to identify which proteins had an experimental structure prior to AlphaFold, and which did not. Even among the selected SwissProt sample, only 8% of proteins had an experimental structure.

The SwissProt subsample also includes a curated literature review for each protein, linking academic papers that have been written about the protein to their SwissProt entries. This allows us to study basic scientific research activity as an outcome. We find statistically and economically significant increases in research activity about previously unsolved proteins compared to their peers that already had experimental structures. New papers about protein function, gene expression, protein-relevant disease, and other protein science categories appear after AlphaFold and are disproportionally focused on proteins where AI revealed its structure for the first time. Our estimates suggest that research on unsolved proteins increased by 23 to 32% relative to previously solved proteins, with effects as large as 80 to 300% by the last year of our sample. We also show that new papers about previously unsolved proteins are much more likely to cite core AlphaFold papers, suggesting that the appearance of AI models are a key mechanism driving this shift in research direction. Together, these results suggest that protein structure determination was likely a key bottleneck in complementary research that has been eased by the introduction of AI tools. Moreover, taken with the complementarity results above, these results suggest that AlphaFold is making exploration of the broader protein space easier, having a sort of “floodlight” effect on protein science. This is consistent with the effects observed in other settings after broad

increases in data access (Nagaraj et al., 2020; Nagaraj, 2021; Tranchero, 2024).

We also investigate whether these new structures led to increased R&D activity, as was initially hoped by observers. We measure R&D using patents, which we are able to link to proteins using a data source called SureChEMBL. SureChEMBL uses text and image-mining to annotate patents, linking references to proteins with their UniProt IDs. In contrast with our basic science results, we see small and only marginally significant upticks in patents referencing previously unsolved proteins. Moreover, our relatively tight standard errors allow us to rule out effect sizes larger than a 13% relative increase. This finding potentially speaks to the role of bottlenecks and complementarities in the research process. Our results suggest that more basic scientific research needs to occur before scientists can make use of these new protein structures in ways that benefits patients and consumers.

Related Research

This paper relates to three literatures. First, it connects to the economics of automation and AI, which studies how new technologies reshape labor markets (Autor, Katz, et al., 1998; Autor, 2015; Humlum, 2021; Boustan et al., 2024; Feigenbaum and Gross, 2024). The task-based framework developed by Autor, Levy, et al. (2003) and Acemoglu and Autor (2011) has been influential in this literature, allowing researchers to study whether technology substitutes for labor in existing tasks, complements workers within those tasks, or creates new tasks altogether. Using the task model, Acemoglu and Restrepo show that automation generates a displacement effect when capital (or AI) takes over tasks previously performed by labor, but technological change can also reinstate labor demand by creating new tasks and reorganizing production (Acemoglu and Restrepo, 2019, 2022; Restrepo, 2024). Recent empirical work on generative AI often finds sizable productivity gains in certain domains,⁴ with especially large gains for less experienced workers (Brynjolfsson et al., 2025; Noy and Zhang, 2023), which could be interpreted as evidence for augmentation rather than one-for-one replacement in the settings studied. Our paper brings this question to a frontier scientific setting in which the potentially automated task—protein structure determination—is unusually well defined.

Second, this paper relates to a growing literature on AI and science. Agrawal et al. (2023, 2024) model AI as a tool that improves scientific discovery by prioritizing search over large hypothesis spaces when experimentation is costly. Ludwig and Mullainathan (2024) similarly emphasize that machine learning can contribute to science not only through prediction, but also by generating hypotheses. Jones (2025) places these ideas in a broader R&D framework, arguing that AI’s impact depends on the share of research tasks it can

⁴ Though not necessarily aggregate, as shown in Humlum and Vestergaard (2026)

perform, the productivity gains on those tasks, and the extent of remaining bottlenecks. Cockburn et al. (2019) and Mullainathan and Rambachan (2025) go further and argue that algorithms may reorganize scientific production itself, including idea generation and theory formation. Relative to this literature, our contribution is empirical and field-specific. Rather than studying AI as a general-purpose research tool, we examine a sharp technological shock in one scientific domain and trace its effects.

Third, and most narrowly, this paper contributes to a fast-growing empirical literature on AlphaFold and its effects on science. Within structural biology, a recurring theme is that AlphaFold complements rather than replaces experiments: Edich et al. (2022) and Kovalevskiy et al. (2024) document rapid adoption while emphasizing that experimental work retains important roles, and Varadi and Velankar (2023) provides case studies of downstream impact, including in drug discovery. Within economics and management, Yu (2026) uses AlphaFold2 as a career shock, showing that more productive structural biologists adopted it faster and that this widened citation polarization. Gans and Mohnen (2026) study the response of experimentalists to AlphaFold and think about the implications for future iterations of the predictive model. Along similar lines, Qian (2026) studies how experimental structural biologists shift which proteins they study, finding that they pivot away from well-predicted proteins. Finally, and closest to our work, Sun et al. (2026) find that a long-running decline in the study of novel proteins halted after AlphaFold2, concentrated among targets with high-confidence predictions.

The remainder of this paper proceeds as follows. Section 2 describes the institutional background and provides a scientific primer. Section 3 describes the data sources and sample construction. Section 4 describes the empirical design and presents results. Section 5 concludes.

2 Institutional background and scientific primer

2.1 Structural biology and the protein folding problem

Structural biology is the study of the form and function of biological macromolecules, especially proteins.⁵ Researchers in this field perform complex experiments to elucidate the three-dimensional shapes of proteins, which are too small to observe under an optical microscope. Traditional experimental approaches such as x-ray crystallography and cryo-electron microscopy (cryo-EM) are time-consuming and expensive. Solving a single protein structure

⁵ Other macromolecules of interest include deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). However, in practice, proteins command the vast majority of scientists’ attention and are the focus of this paper.

can take months to years and may cost on the order of \$100,000 or more, depending on the method and difficulty of the target (Sullivan et al., 2017).⁶ Since the field’s development in the 1970s, researchers have solved and publicly deposited around 130,000 unique protein structures.

Protein structures are important because they reveal how proteins function inside the cell. Structural maps allow researchers to see how the protein folds, where it binds to other molecules, and how mutations might alter its behavior. Structural information also enables advances in disease biology and drug design. For example, structural information about HIV-1 protease was critical for developing the earliest HIV treatments which blocked the activity of this protein (Erickson et al., 1990). More recently, the rapid determination of the SARS-CoV-2 spike glycoprotein structure enabled the rational design of stabilized spike antigens used in mRNA Covid-19 vaccines (Wrapp et al., 2020).

Proteins are composed of chains of small molecules known as amino acids. In almost all organisms, there are 20 standard amino acids used to build proteins. A protein’s three-dimensional shape arises from the physical and chemical interactions among these amino acids, which cause the chain to fold into a specific structure. The amino acid sequence of a protein is coded directly from DNA. As a result, determining a protein’s amino acid sequence is comparatively straightforward: it can be inferred directly from genomic sequencing data.⁷ By the completion of the Human Genome Project in 2003, essentially the full set of human protein-coding sequences had been identified (International Human Genome Sequencing Consortium, 2004). The advent of next-generation sequencing technologies in the mid-2000s dramatically reduced the cost of DNA sequencing and enabled large-scale sequencing of other organisms, viruses, and bacteria (Metzker, 2010). Today, public databases contain sequences for hundreds of millions of proteins across all domains of life (The UniProt Consortium, 2025). Despite this explosion in sequence information, only a tiny fraction (less than 0.1%) of known proteins have experimentally determined three-dimensional structures.

Given that amino acid sequences are comparatively cheap and easy to deduce compared to experimental structures, it is not surprising that scientists have long been interested in trying to predict how proteins will fold based on their amino acid sequence alone. Indeed, the so-called *protein folding problem*—determining a protein’s three-dimensional structure from its amino acid sequence—has been a central open question in structural biology for more than half a century (Anfinsen, 1973). In practice, however, the mapping from sequence

⁶ Although cryo-EM has accelerated structure determination for many large proteins, it requires multi-million dollar instrumentation and historically produced lower-resolution structures than crystallography, particularly prior to the so-called “resolution revolution” in the 2010s (Nogales, 2016).

⁷ Each amino acid is coded by three DNA base pairs. For example, the DNA sequence ATG codes for the amino acid methionine.

to structure is extraordinarily difficult due to the vast number of possible conformations a protein can take.⁸

2.2 Structure prediction and AlphaFold

Despite the challenges associated with structure prediction, computational biologists launched the Critical Assessment of protein Structure Prediction (CASP) in 1994 as a recurring, blinded evaluation of prediction methods (Moult et al., 1995). Every two years, groups could submit predictions of protein structures whose experimental coordinates had been determined but not yet released publicly. Once the experimental structures were made available, predictions were scored by organizers against the experimentally determined ground truth. The Global Distance Test-Total Score (GDT_TS) roughly measures the fraction of amino acids that are placed “close enough” to the experimental ground truth (Zemla, 2003), and CASP assessors had long treated a GDT_TS score of 90 as equivalent to experimental accuracy (Kryshtafovych, Schwede, et al., 2021).

While the technical details of these prediction models are beyond the scope of this paper, one basic point is important. Modern protein structure prediction models make use of information from evolution. By comparing related protein sequences across many organisms, they can detect statistical patterns that reveal which amino acids are likely to be close together in the folded structure. These patterns help constrain the set of plausible three-dimensional shapes.⁹ Importantly, AlphaFold2 does *not* depend primarily on having a closely related experimental structure available. Instead, it can infer structure directly from the amino acid sequence, together with evolutionary information from related sequences.¹⁰ As a result, AlphaFold can perform well even for proteins without a similar experimental structure in the training data (what economists might call “out of sample”), as long as a sufficient number of similar protein sequences are available.

Progress at CASP was slow, with winning teams’ average GDT_TS scores rarely exceeding 40 through 2016. A discrete shift occurred at the 13th edition of CASP in 2018, where Google DeepMind’s AlphaFold system appeared for the first time and substantially outperformed competing approaches, achieving an average GDT_TS score of nearly 60 across all targets. When DeepMind returned in 2020 with AlphaFold2 (Jumper et al., 2021), they astonished

⁸ In 1969, biophysicist Cyrus Levinthal put this into perspective with “Levinthal’s Paradox,” which states that a small protein of 100 amino acids has about 10^{47} possible conformations. Even if proteins could sample conformations at the speed of molecular vibration— 10^{13} per second—it would take longer than the age of the universe to sample them all (Levinthal, 1969).

⁹ For interested readers, Ball (2024) provides a non-technical summary.

¹⁰ Consistent with this, Jumper et al. (2021) show that AlphaFold2 performs nearly as well even when experimental structure information is excluded.

CASP evaluators, participants, and observers when they achieved an average GDT_TS score of nearly 90 across all competition targets. This led John Moult, the founder of CASP, to say “in some sense, the [protein folding] problem is solved” (Callaway, 2020). Although DeepMind may have been expected to win, the progress they made—and the declaration that the protein folding problem had been solved—was unexpected by the structural biology community. Coverage of CASP 14 in 2020 describes the breakthrough as “remarkable” and “startling” (Callaway, 2020; Kryshchuk, Schwede, et al., 2021), with some observers describing an atmosphere of shock and denial (Ball, 2024).

The news of DeepMind’s success at CASP was released in November of 2020. By July of 2021, they had released the underlying code for AlphaFold2. Simultaneously, they ran the algorithm on hundreds of thousands of known amino acid sequences, and uploaded the predicted protein structures directly into a freely accessible database.¹¹ This made access to the structure predictions frictionless for researchers (Varadi and Velankar, 2023).

It is important to highlight that structure prediction has continued to advance since AlphaFold2. DeepMind first extended the model to protein complexes with AlphaFold-Multimer (Evans et al., 2022), and in 2023 released AlphaFold3 (Abramson et al., 2024) which targeted further weaknesses of AlphaFold2 (see Appendix B for more discussion of AlphaFold2’s weaknesses). Other labs have since provided similar models, though none have been as freely distributed as AlphaFold2.¹² Impacts we observe in the later years are likely due to a combination of different structure prediction technologies.

2.3 Protein structures and downstream innovation

Much of the excitement around AlphaFold was focused on the downstream scientific advances it might enable or encourage. Seeing a protein structure for the first time can allow researchers to develop new hypotheses about its biological function, its interactions with other proteins, or its role in disease. Scientists can then develop experiments to test these new hypotheses, increasing our knowledge and understanding of these proteins beyond just their three-dimensional structure.

These new insights may also translate to applied research and development. Particular

¹¹ In the following months, DeepMind expanded the number of structures covered in this database. We discuss this more in Section 3.

¹² It seems likely that AlphaFold2’s impact depends on an unusual degree of openness: freely available code, a browser-based interface (ColabFold), and, above all, a precomputed database of over 200 million predicted structures with zero access restrictions. AlphaFold3, by contrast, was initially available only through a rate-limited web server. Its code was released in late 2024, under a non-commercial license with model weights gated to academic users. Fully open reproductions have since emerged, but no subsequent models have been distributed with anything approaching AlphaFold2’s freely downloadable database of precomputed structures.

attention has been paid to drug discovery. Most drugs are small molecules that function by binding to a protein target, inhibiting or modulating its activity (Santos et al., 2017). Returning to an example from above, saquinavir is a drug that targets HIV-1 protease. HIV-1 protease is an enzyme that the HIV virus uses to replicate inside of cells. Thus, blocking this enzyme interrupts the viral life cycle at a critical step (Erickson et al., 1990).

In the case of HIV-1 protease inhibitors, understanding the protein’s structure was critical in developing the drug. Experimental structures published by structural biologists revealed a well-defined binding site (Navia et al., 1989; Wlodawer et al., 1989). Researchers used the shape of this binding site to design molecules that would fit snugly into it and block the enzyme’s activity—an approach known as structure-based drug design.

In addition to small-molecule drugs, protein structures can facilitate the design of antibodies and other biologics, diagnostics and chemical probes, and vaccines. As referenced above, the Covid-19 vaccine development relied heavily on experimentally solved structures of the SARS-CoV-2 spike protein (Wrapp et al., 2020). Beyond medical innovation, protein structures can aid in the development of agrochemicals, which typically target non-human proteins (such as weed or insect enzymes). It can also assist with the modification or design of new enzymes used for green chemistry, manufacturing, food, and biofuels.

3 Data sources and construction

In this section, we describe our data and how we construct our main analysis samples. For more detail, please see Appendix A.

3.1 The Protein Data Bank (PDB)

Our data source for experimental structure discoveries is the Protein Data Bank (PDB). Founded in 1972, the PDB is an online repository of biological macromolecular structures that are deposited by structural biologists around the world. Since the 1990s, deposition in the PDB has been mandatory for researchers seeking to publish academic papers (Strasser, 2019). We downloaded our data in April 2026, and at that time, there were over 250,000 structure deposits, 95% of which are proteins. The primary information that the PDB is designed to host is the three-dimensional molecular coordinates that describe the proposed protein structure. They also collect and categorize a rich set of metadata for each project.

We observe three key dates that help us re-construct the timeline of each protein structure. First is the collection date, which is the date that scientists collect their experimental data from a synchrotron x-ray facility. Second is the deposit date, which is a timestamp for

when the project details were first uploaded to the PDB servers. At this point, the data are held confidentially while the authors typically go through the journal submission and revision process. Finally, the data are made public on the release date, usually coordinated with journal publication, or otherwise a maximum of one year after the deposit date. The median time between collection and deposit is 11 months, and the median time between deposit and release is 5 months.

In addition to these dates, we also gather experimental details from the PDB. This includes the broad experimental method that the researchers use to determine the structure (for example, x-ray crystallography or cryo-EM). It also includes the computational techniques they use to turn their data into three-dimensional structures, the software and hardware they used, and any existing protein structures they used as templates.

3.2 AlphaFold Protein Structure Database

When Google DeepMind first used AlphaFold to predict protein structures at scale, they partnered with the European Molecular Biology Laboratory’s European Bioinformatics Institute (EMBL-EBI) to host the predicted structure data for public use. In July 2021, DeepMind released the first 365,000 predicted protein structures as well as the open-source code for the prediction model. These proteins were selected for their biological relevance, and included virtually all human proteins. In December 2021, they released an additional 560,000 structures, and in July 2022, they released a prediction for every known protein (200 million+ structures). We treat July 2021 as the treatment date, because this is when researchers gained access to predicted structures for the first time. We downloaded our version of the data in May 2026.

This dataset, shortened to AlphaFold DB, is our primary source of data for AI-predicted structure details. For each structure, AlphaFold DB provides a confidence score known as average predicted Local Distance Difference Test (pLDDT). pLDDT represents how confident the model is in the predicted position of each amino acid in a protein structure, and average pLDDT takes the average across all amino acids in the structure. Multiple studies have shown that pLDDT is strongly correlated with actual prediction accuracy, using proteins that have an experimental structure that was not used in the model training (Jumper et al., 2021; Tunyasuvunakool et al., 2021; Edmunds et al., 2024).

3.3 UniProt and SwissProt

In order to study the impact of AlphaFold on broader areas of basic protein science, we rely on the Universal Protein Resource (UniProt), a collaboration between the EMBL-EBI,

the Swiss Institute of Bioinformatics, and the Protein Information Resource. UniProt is a standardized database of protein details, including literature citations. While all of UniProt contains over 200 million protein entries, we focus on a subset of UniProt called SwissProt, a manually curated selection of just over 570,000 proteins that are particularly biologically relevant. There is a variety of criteria used to determine which proteins are included in SwissProt. It includes almost all proteins from humans and other key model organisms or major pathogens.¹³ It also includes proteins known to be relevant to medicine and disease, as well as proteins with strong experimental evidence on their function or activity, structure, subcellular location, or other traits. We downloaded our version of the data in June 2026.

An advantage of using the literature citations curated by SwissProt (as opposed to trying to match against all publications using protein keywords) is that the citations are extremely high-quality and take advantage of curator expertise. Trained biologists read the literature and screen papers based on several criteria: Is the protein the central focus of the paper?; Is the evidence provided in the paper strong?; Is the paper redundant?; Is the paper a review article or comment? Only a small share of papers that might be flagged by keywords meet these standards (Poux et al., 2017). Thus, these references reflect meaningful new knowledge. In addition, SwissProt pulls citations directly from other annotated databases.¹⁴

Importantly, SwissProt also provides paper categorizations which describes the content of the paper in distinct categories. These include function, process, disease, sequence, family, etc. One of the categories is structure, which typically denotes a link between the paper and a PDB deposit. Since we are interested in studying the effect of AlphaFold on research *beyond* experimental structural biology, we drop all structure papers.

An example. For years, biochemist Andrea Pauli had been trying to understand how sperm and egg cells fuse, working with zebrafish.¹⁵ Her lab had found a protein on the surface of zebrafish egg cells known as Bouncer (UniProt ID P0DPQ9) and showed it was essential for fertilization. But they still did not understand how Bouncer recognized sperm cells. AlphaFold revealed both the structure of Bouncer and of a different protein, known as Tmem81 (UniProt ID B8JI67), whose role in reproduction had not been known. This allowed Pauli and team to hypothesize that 3 sperm proteins—including Tmem81—formed a complex that recognized Bouncer. They conducted additional experiments to validate this hypothesis, and published their findings in *Cell* (Deneke et al., 2024).

Deneke et al. (2024) is one of the papers in the SwissProt curated literature. The paper

¹³ It includes the full proteome of mice, Arabidopsis, yeast, E. coli, HIV, and influenza, among others.

¹⁴ Including the PDB, RefSeqs, Mouse Genome Informatics, and others.

¹⁵ This example is taken from a five-year retrospective of AlphaFold’s impact published in *Nature*, (Callaway, 2025).

has been manually linked by expert curators to eight different SwissProt proteins, including Tmem81 and Bouncer. Thus in our data, each of these eight proteins gets a new linked publication. Alternatively, if we do not want to count the same paper multiple times, we can assign each linked protein a *fraction* of the paper, splitting the paper evenly among all linked proteins. In this example, since Deneke et al. (2024) is linked to eight proteins, each protein would get 1/8th of the paper.

3.4 OpenAlex

OpenAlex is an open-source database of scholarly works covering hundreds of millions of academic publications (Priem et al., 2022). We use OpenAlex to supplement our SwissProt curated papers with reference lists. This allows us to see which SwissProt papers are citing core AlphaFold references. We focus on the three references that EMBL suggests users cite when using AlphaFold DB: Jumper et al. (2021), Varadi, Anyango, et al. (2022), and Varadi, Bertoni, et al. (2024).

3.5 SureChEMBL

SureChEMBL is a public database maintained by EMBL-EBI that organizes the chemical information disclosed in the patent literature (Papadatos et al., 2016). It is updated frequently; we downloaded our data in May 2026.¹⁶ It is assembled via a text- and image-mining pipeline that scans the full text of patents from the major patent offices¹⁷ going back as far as 1920 to identify and extract chemical compounds. Once the patents linked to chemicals have been identified, SureChEMBL further annotates them, pulling out references to diseases, genes, and—critically for our purposes—proteins. SureChEMBL takes a dictionary-based approach, standardizing protein references and assigning a UniProt ID. Thus, we can link proteins to the patents that mention them. Of the over 44 million patents indexed by SureChEMBL, 5.5 million link to at least one UniProt ID. In total, over 100,000 unique UniProt IDs are contained in the SureChEMBL patent data.

To focus more narrowly on medical innovation, we also flag patents that are related to medical technology. We do this by using the WIPO IPC-Technology Concordance, which maps patent classification codes to 35 technology fields (Schmoch, 2008). We code any patent containing an International Patent Classification (IPC) code in the “pharmaceutical,” “biotechnology,” or “medical technology” fields as a medical patent.

¹⁶ We use the version 2.0 database first released in May 2025.

¹⁷ These include the US Patent Office, European Patent Office, World Intellectual Property Organization, Japan Patent Office, and China National Intellectual Property Administration.

An important limitation of the SureChEMBL data is that patents will only be annotated with UniProt IDs if they contain a chemical compound. This is not a limiting factor for small-molecule drugs targeting a protein. In this case, the drug itself is a chemical compound and will thus lead to the patent’s inclusion, allowing for the protein target to be annotated. However, for other protein-related innovations, coverage may be thinner. For example, biologics are drugs that come from living organisms, like proteins and genes. They are not chemical compounds. If the patent disclosing a biologic does not contain any references to chemical compounds, then the patent will not be included in our data. However, [Appendix Table 1](#) and [Appendix Table 2](#) show the most common four-digit IPCR codes found in SureChEMBL patents linked to a UniProt ID. These codes suggest that many medical innovations beyond small-molecule drugs *are* being included in our data. The C12N (microorganisms, enzymes, genetic engineering) and C07K (peptides) subclasses shows up as the third and fourth most common IPCR codes in both tables; these are exactly the subclasses where biologics would appear.

3.6 Survey of structural biologists

To complement the administrative data from structural biology and related fields, we also run a survey of researchers active in experimental structural biology. We target principal investigators—researchers who direct labs—who have deposited experimental structures in the PDB during our sample period. We sent out just over 4,000 surveys and received 427 complete responses, for a response rate of 11%. We ask respondents a series of questions about how AlphaFold has impacted their field today, as well as about their expectations for the future. More details about the survey can be found in [Appendix A](#), and the full survey instrument can be found in [Appendix D](#).

3.7 Data construction

Experimental structure data

Our goal is to build a structure-level dataset that will measure the rate of research in experimental structure determination. We start with the universe of PDB structure deposits from 1972 to April 2026—a total of 252,448 structures. We then make a series of restrictions.

First, we drop duplicate structures that are part of group deposits. Occasionally, researchers will deposit hundreds (or even thousands) of near-identical structures by the same team on the same day.¹⁸ Because these submissions do not represent a proportional increase

¹⁸ Research teams may do this for a few reasons. One example is a lab may solve the same protein

in distinct scientific effort, we retain one deposit in these groups. In some cases, these group deposits are explicitly marked, making this easy. In other cases, we infer group deposits if the protein structures are (1) solved by the identical authors; (2) deposited on the identical day; and (3) have the identical amino acid sequence. Dropping these duplicates leaves us with 181,365 structures.

Second, we drop structures from the SARS-CoV-2 (Covid-19) virus, in an effort to normalize activity around the pandemic. This only impacts 3,002 structures, leaving us with 178,363 structures. Third, we drop non-protein structures (primarily DNA and RNA structures). This leaves us with 173,225 structures.

We focus on structures that were deposited between 2017 and the first quarter of 2025. In this time frame, we have a total of 71,133 structures. We drop the last 12 months of our data, which runs through the end of the first quarter of 2026. We do this because of the 12-month release lag for experimental structures. During the last 12 months of our data coverage, additional structures will be deposited but unreleased (and thus invisible to us), leading us to under-count structures in that time period. Table 1 presents the summary statistics for our analysis sample.

We also merge on information about the predicted analogues of experimental PDB structures from AlphaFold DB using UniProt IDs. As shown in Table 1, about 16% of PDB structures in our sample do not have a predicted analog. This can arise for several reasons. AlphaFold did not predict proteins with 15 or fewer amino acids, proteins with more than 2700 amino acids, or viral proteins,¹⁹ so these UniProt IDs do not have predictions. Moreover, some proteins deposited in the PDB are not mappable to UniProt, such as engineered proteins or antibodies that are patient-specific.

Spillover panel

Our goal is to build a panel dataset that tracks additional research, beyond experimental structure determination, using data from the SwissProt curated literature and SureChEMBL annotated patents. We start with the list of 570,829 SwissProt-indexed proteins. We drop any proteins that do not have a predicted structure in AlphaFold DB, which occurs for the same reasons discussed above. This comprises four percent of the SwissProt sample.²⁰ We

repeatedly with different bound compounds. Each complex is scientifically useful and gets its own PDB deposits, but much of the experimental setup is shared across deposits.

¹⁹ Large proteins were omitted for computational reasons, and viral proteins were omitted due to the issue of polyproteins.

²⁰ Note that this implies that we are dropping any Covid-19 proteins due to viral restrictions. In general, other databases have sprung up to include predictions of these omitted proteins (such as the Big Fantastic Viral Database (Kim et al., 2025)). Thus, it is uncertain to what extent these proteins have structures available, and we view it as safest to drop these.

are left with a sample of 546,609 proteins.

We then attach the SwissProt curated publications to their associated proteins. We drop any publications that are categorized as “structure” publications, as these typically correspond to a PDB deposit, and we are aiming to measure research that occurs beyond structure determination. For every remaining paper, we observe the protein it is linked to and its publication year. Some papers are linked to multiple proteins. We handle this in different ways. One way is to count every paper-protein link. Another is to assign fractional shares of papers to proteins. We count the number of papers (and fractional papers) linked to each protein from 2017 to 2025, creating a panel.

Next, we link SureChEMBL patents by their curated UniProt ID(s). Since the same invention may have multiple applications across multiple patent agencies, we collapse patents down to the family level, assigning the earliest publication date. We count the number of unique patents linked to each SwissProt protein in every year from 2017 to 2025.

Finally, we link experimental structure data from the full PDB, again using UniProt IDs. This allows us to compute when a structure was first solved, and to measure whether a given protein had an experimental structure that was publicly released prior to AlphaFold.

Table 2 presents the summary statistics for the panel. 92% of the SwissProt proteins were unsolved prior to AlphaFold. For paper and patent counts, we show full counts all the way through 2020, stopping before AlphaFold was introduced. The average protein was linked to 3.6 papers and was mentioned in 9.9 patents prior to AlphaFold’s release, 6.0 of which were medical in nature. We highlight that these counts are skewed, with a small number of proteins receiving a disproportionate share of study and innovation in the pre-AlphaFold period. For both papers and patents, only 17% of proteins are ever linked prior to 2021. The average AlphaFold confidence score for the predicted structures is 87.5, which DeepMind classifies as “high.”

We end our panel in 2025. [Appendix Figure 1](#) and [Appendix Figure 2](#) show the histograms of total linked papers and patents from 2017 to 2025. Because the SwissProt-linked papers are curated in many cases by hand, there is a lag between when papers are published and when they are linked. Thus, we observe a lower number of linked papers in 2025 (and to a lesser extent, 2024). These totals will likely increase over time as curators backfill previously published papers.

4 Empirical strategy and results

4.1 Structural biology

Our first set of results focus on how AlphaFold has impacted the field of structural biology. Has AlphaFold served as a substitute for experimental structure determination, or as a complement to it? If the AI predictions are a good substitute for experimental structures, do we see a decline in effort and output from traditional structural biology methods? Figure 2 shows simple count statistics over time of experimental work. Panel (a) shows the count of experimentally determined structures in the PDB. We index proteins by their deposit date, which is the date they were uploaded to the PDB (but not publicly released). We drop the last 12 months of data, since authors have up to a year to publicly release their deposits. We see no evidence that the rate of experimental structure solving has slowed post-AlphaFold. In fact, researchers appear to be depositing structures at a slightly *higher* rate after July 2021. The results are nearly identical if we add back in Covid-19 structures (see [Appendix Figure 3](#)). If we plot all protein structures with no sample restrictions, the results are qualitatively similar, though slightly noisier due to the large group deposits (see [Appendix Figure 4](#)).

Moreover, it is not the case that scientists are merely finishing up work that they started prior to AlphaFold’s release. When we investigate collection dates—the date that scientists collected their experimental data—we find in [Appendix Figure 5](#) that over 75% of deposits near the end of our sample window had collection dates after AlphaFold’s release. This implies that researchers are continuing to start new experimental projects.

There also appears to be continued interest in these experimentally solved structures. Panels (b) and (c) show that these structures continue to publish in journals and in “top journals” (defined as *Cell*, *Nature*, and *Science*) at similar rates before and after AlphaFold.

Why hasn’t AlphaFold displaced experimental structural biology?

Given the impressive capabilities of AlphaFold and similar AI tools, why are scientists continuing to carry out expensive and time-consuming experiments? One potential explanation might be simple inertia. Experimentalists are trained to use these methods, so might continue doing experimental work as long as possible even if their output becomes obsolete. However, top-ranked general interest journals with editors that are incentivized to publish frontier research are still showcasing experimental results. Researcher incentives may play a role, but the evidence suggests that it is not the primary explanation. Instead, we propose and test three alternative explanations for the lack of displacement of experimental methods

in the face of capable AI tools: 1) imperfect and unpredictable reliability of AI predictions, 2) adaptation in experimental project selection, and 3) complementarity between AI predictions and experimental methods. We find evidence supporting all three of these hypotheses as joint explanations for the trends we document.

4.1.1 Reliability: Experiments confirm or correct AlphaFold predictions

A first explanation, argued persuasively by many structural biologists, is that while predictions are useful, they are not perfect substitutes for experimental structures. One reason relates to the accuracy of the predicted structures. While highly accurate on average, Terwilliger et al. (2024) found in a careful comparison of around 100 experimental versus predicted structures that some amino acids can be misplaced, even when the confidence scores for that amino acid are very high—they estimate that about 10% of “very confidently” placed amino acids are in fact meaningfully misplaced.

Our survey of structural biologists supports this view. As reported in Figure 3, when asked “Do you trust AlphaFold2 or other protein structure predictions enough to skip experimental validation?” 29% report that they never trust the predictions and 42% report that they usually do not trust the predictions. Only 6% report that they trust the predictions always or most of the time. This lack of confidence might change over time as the AI tools become more capable, but it is clear that the current majority belief, five years after AlphaFold2, is that experimental insights are still valuable due to imperfect reliability of AI predictions.

4.1.2 Adaptation: Experimentalists pivot toward comparative advantage

Another possibility is that AlphaFold is a substitute for experimental protein structures more narrowly, among the structures that it is most capable of predicting. If this is the case, we would expect to see experimentalists pivot away from these proteins, toward proteins where they retain a comparative advantage. To get at this question directly, we ask our survey respondents: “Agree or disagree: AlphaFold2 and related advances in protein structure prediction have caused me to shift my research focus toward experimentally solving proteins that are predicted poorly by AI methods.”

Panel (a) of Figure 4 reports the responses and suggests a fair amount of disagreement on this topic. About 22% of respondents strongly or somewhat disagree with this statement, and 26% neither agree nor disagree, suggesting that about half of the community is indifferent to the rise of AlphaFold when it comes to project selection. On the other hand, the remaining 52% of respondents somewhat or strongly agree that they have shifted their research in

response to AlphaFold2, focusing more on proteins that AlphaFold cannot predict well.

We also attempt to measure this directly in the PDB data. pLDDT confidence scores provide one proxy for AlphaFold’s predictive capability, but they do not tell the entire story. AlphaFold struggles in other ways not captured by confidence, including with multi-entity structures, proteins bound to other compounds, mutated proteins, and proteins in non-standard conformations (Terwilliger et al., 2024). Thus, we develop a classification of proteins as being “easy” or “hard” for AlphaFold to solve that takes these different factors into account, as described in detail in Appendix B.

Panel (b) of Figure 4 shows the share of easy and hard proteins over time. If there is a shift in the data, it is difficult to detect. Even before AlphaFold, there appears to be a slight decline in structures that are easy for AlphaFold, a trend which continues after 2021. After AlphaFold2, there appears to be small uptick in experimental structure deposits that would be difficult for AlphaFold, but it is very slight.

As a simpler exercise, we also plot the mean confidence score over time. If researchers pivot toward structures that AlphaFold is less confident in, this would show up as lower average confidence scores among experimental structures. However, Appendix Figure 6 suggests that this is not the case: the average predicted confidence of experimentally determined structures remains fairly constant throughout our sample period. Comparing the pre- and post-AlphaFold mean suggests confidence scores dropped by one point, but given that the standard deviation in confidence scores in the PDB sample is over ten, this is a very small effect. In unreported analyses, we repeat this exercise using the share of proteins with confidence scores above 90, 80, and 70 as the outcome, finding very similar results.²¹ Taken together with our survey results, we interpret this as suggestive evidence that some degree of pivoting is happening, but caution that the results do not appear to be conclusive.

4.1.3 Complementarity: AlphaFold facilitates more efficient experimentation

If AlphaFold is not serving as a substitute for experimentally determined structures, is it complementing this work? There have been several accounts of predicted structures enabling researchers to solve experimental structures that had previously eluded them (Kryshtafovych, Moulton, et al., 2021). To probe this question, we need to introduce two new scientific concepts: molecular replacement and protein sequence homology.

²¹ It is important to note that we drop PDB proteins that do not have a predicted counterpart, rather than imputing them as zero. Researchers might be able to find predictions for these proteins in other databases (Kim et al., 2025), or might be able to fold them themselves using ColabFold. Thus, it is unclear how to correctly impute confidence for these proteins. This likely explains a large share of our disagreement with Qian (2026), who imputes zeroes and finds that *within* narrow protein classes, structural biologists do meaningfully shift away from very well-predicted proteins (> 90 average pLDDT scores).

Molecular replacement. The most common experimental technique in our sample is called x-ray crystallography, with over 70% of the structures in our sample using this technique. As outlined in Hill and Stein (2025a), this technique broadly consists of three steps: First, researchers crystallize proteins. Second, they take the crystals to specialized synchrotron facilities and beam them with x-rays, generating experimental data known as a diffraction pattern. Third, they use the experimental data to reverse-engineer the structure that generated it.

This third step can be performed in a variety of ways, but the most common method is known as molecular replacement (MR). Almost 90% of experimental x-ray structures in our sample employ this technique. The core idea behind MR is that it uses a similar known protein structure as the starting point or “template” for model building. As discussed by Kim (2026), this makes structure solving by MR easier and faster than by other techniques, especially since the development of specialized software in the early 2000s (McCoy et al., 2007). However, MR can only be employed if a similar structure already exists.

Prior to 2021 and the introduction of AlphaFold, this meant that another similar protein must have been solved experimentally in order for researchers to use MR. However, after the development of AlphaFold and the mass deposition of predicted structures in July 2021, scientists quickly realized that *predicted* structures could also be used as the starting structures in MR.

Defining proteins with sequence homologs. What does it mean for an unsolved protein to have a “similar enough” experimental structure? The general rule is that if a protein shares at least 30% of its amino acid sequence with another protein, that second protein can be used as a template for MR (Phenix, n.d.; Kim, 2026). Thus, for every protein in our sample, we perform the following computations:

1. We find the pool of all experimentally solved proteins in the PDB that were released prior to the focal protein’s deposit date. These proteins go all the way back to the 1970s and represent all possible (public) structures that a researcher could have used as their starting structure.
2. We compare the focal protein’s amino acid sequence to that of every protein in the pool, and calculate the sequence similarity. We are able to do this using specially designed software for this task known as MMseqs2 (Steinegger and Söding, 2017).
3. We find the focal protein’s nearest neighbor—the protein in the pool with the highest sequence similarity. This can range from 0 to 100, with the distribution shown in

[Appendix Figure 7](#). We call the percent similarity between these two proteins the sequence homology score. Hereafter, we will shorten this to “homology score.”

We use this continuous homology score and an indicator for “has homolog” (which equals one if the homology score is greater than or equal to 30) in our subsequent analysis.

Testing for complementarity. We begin by investigating the relationship between homology score and the use of MR, before and after AlphaFold. We restrict to structures solved via x-ray crystallography, since MR is only possible for these structures. Prior to AlphaFold’s introduction, we expect to see an increasing probability of using MR as the homology score increases. However, if AlphaFold truly does expand the set of proteins that are amenable to MR, we would expect to see this relationship flatten in the post-period.

To test this, we split the PDB sample into five roughly evenly sized groups based on homology scores (as [Appendix Figure 7](#) shows, there is significant mass at 0 and 100). The first group contains all proteins with a homology score of exactly zero. The second group contains all proteins with a score between 0 and 33, the third all proteins with a score between 33 and 66, and the fourth all proteins with a score between 66 and 100. The final group contains all proteins with a score of exactly 100. We then define a *Post* indicator, which equals one if the protein was deposited after July 22, 2021, the date that the first wave of AlphaFold predicted structures were publicly released. We index structure deposits by i and homology groups by $g \in \mathcal{G} = \{0, 33, 66, 99, 100\}$ and estimate:

$$MR_i = \sum_{g \in \mathcal{G}} \beta_g \cdot D_{ig} + \sum_{g \in \mathcal{G}} \gamma_g \cdot D_{ig} \cdot Post_i + \varepsilon_i \quad (1)$$

where D_{ig} is an indicator equaling one if protein i is in group g .

Figure 6 presents the results. The blue circles show the trend in the pre-period, plotting β_g for each group. For structures with a homology score of zero, we see just under 50% of them using MR. This rises steeply as homology increases: in the $(0, 33]$ group this rises to over 70%, and increases to 90% in subsequent groups.

The red triangles plot the sum of $(\beta_g + \gamma_g)$ coefficients and the 95% confidence interval of γ_g . Across all homology bins, MR becomes more common in the post-period. The increase is largest at low levels of sequence similarity—precisely where pre-period usage was relatively low. In the zero-homology group, MR usage rises by 26 percentage points. In the $(0, 33]$ group, it increases by 14 percentage points. In contrast, for high-similarity structures—where MR was already used in over 90% of cases pre-AlphaFold—increases are mechanically constrained and range from five to seven percentage points. Overall, the relationship between homology and MR usage is substantially flatter in the post-period compared to the pre-

period, suggesting that AlphaFold complements experimental structure determination for low-homology proteins by making them amenable to MR.

To further hone in on this theory, we perform two additional tests that take advantage of the sharp timing of AlphaFold’s release. We begin by dividing the sample into two groups: “structures with homolog,” defined as experimental structures where the nearest experimental neighbor had at least 30% sequence similarity, and “structures without homolog,” defined as the converse. Then, we study how the two groups evolve over time. Letting i again index structure deposits and q index quarter of deposit, we estimate:

$$Y_i = \alpha + \lambda Homolog_i + \sum_{q \neq 2021Q2} \delta_q \cdot D_{iq} + \sum_{q \neq 2021Q2} \theta_q \cdot D_{iq} \cdot Homolog_i + \varepsilon_i \quad (2)$$

where $Homolog_i$ is an indicator for whether structure deposition i has an experimental homolog at the time of deposit and D_{iq} is an indicator for whether i was deposited in quarter q .

In Figure 7, we let the outcome be the use of MR, and restrict to structures that were solved using x-ray crystallography. Panel (a) plots the rates of MR usage by the two groups (equivalent to $\alpha + \delta_q$ for structures without a homolog, and $\alpha + \lambda + \delta_q + \theta_q$ for structures with a homolog). Panel (b) plots the difference ($\lambda + \theta_q$) and the 95% confidence interval of θ_q . We can see that the gap in MR usage between structures with and without a homolog only begins to close after the arrival of AlphaFold. At baseline, the difference is about 40 percentage points. By the end of our sample, it has fallen below ten percentage points. The timing suggests that AlphaFold is the causal mechanism behind the closing of the MR gap.

However, our data lets us probe this even more closely. When researchers use MR, they are encouraged to report the structure they used as a template and its source in the PDB. About 90% of MR structures in our sample reported a starting structure. Prior to AlphaFold, over 98% of reported starting models came from the PDB. However, post-AlphaFold, about 12% of structures reported using an AlphaFold starting structure.

Figure 8 estimates Equation 2 using an indicator for an AlphaFold starting structure as the outcome. We restrict to structures solved via x-ray crystallography, using molecular replacement, and citing any starting structure. Mechanically, in the pre-period, neither group cites any AlphaFold structures.²² However, in the post-period we rapidly see that researchers are using AlphaFold predicted structures as their starting models. This use is far more concentrated among structures without experimental homologs. By the end of our sample period, structures without homologs are using AlphaFold predicted structures as their

²² The tiny uptick in using an AlphaFold predicted structure in Q2 of 2021 comes from four experimental structures, and likely represents either a typo or private access to AlphaFold predictions.

starting models over 60% of the time, compared to 15% for structures with homologs. This is consistent with AlphaFold serving as a complement to continued experimental structure determination, especially for structures that are more difficult to solve due to their novelty.²³

Structural biology going forward

One potential weakness of our analysis thus far is that five years may simply not be enough time to observe meaningful impacts on the field of structural biology. However, as part of our survey, we also asked structural biologists about their expectations for the field in the future. More specifically, we asked respondents to tell us the number of experimental structures their lab solves annually and the share of their lab’s time that is devoted to experimental structure determination. For both questions, we asked about the pre-AlphaFold era (defined as 2017 to 2020), the post AlphaFold era (defined as 2021 to 2026), and the future (defined as 2031). The results are shown in Figure 5.

Panel (a) shows that our survey respondents report a slight uptick on average in structures solved per year before and after AlphaFold (from 6.8 to 7.7). This 13% increase is nearly identical to the increase in PDB deposits that we observe in Figure 2. By 2031, our survey respondents expect this number to increase by another 30% to about 10 structures per year.

Panel (b) shows the share of lab time devoted to experimental structure determination. Despite the increase in structures, the share of lab time has actually decreased slightly, from about 47% before AlphaFold to 40% after. This likely speaks to the gains in productivity that arise from using predicted structures as a complement to experimental structures, as discussed above. By 2031, survey respondents expect to spend slightly less time than they do today, estimating that about 36% of their lab’s time will be spent on experimental structure determination. Taken together, these results suggest that even a decade after AlphaFold2’s release, scientists in this field still expect to spend significant time experimentally solving protein structures. Moreover, they expect to solve *more* of these experimental structures than they are solving today.

4.2 Spillovers to related fields of basic research

Empirical strategy

We now consider the broader impact of AlphaFold on related areas of science and downstream applied R&D. Here we exploit the fact that AlphaFold predictions represented a

²³ The scientific literature (e.g., Akdel et al. (2022)) suggests that a similar phenomenon happens for structures that are determined using cryo-electron microscopy (the other major experimental technique apart from x-ray crystallography, comprising 22% of our sample), through a process called docking. However, this is harder to trace in the data.

shock of structural information about some—but not all—proteins. By mid-2021 when AlphaFold predictions were first posted and the model was released publicly, some proteins had already been solved by experimental methods. As shown in Table 2, about 8% had an experimentally solved structure in the PDB. More broadly, about 40% had the experimentally solved structure of a close homolog. AlphaFold therefore represented a sudden and unexpected endowment of new structural knowledge for unsolved proteins relative to solved proteins. Figure 9 illustrates this logic. We show two examples. Prior to 2021, UniProt ID P69905 had an experimental structure already deposited in the PDB. However, UniProt ID Q9NPJ8 did not. After 2021, both of these proteins had a predicted structure. For P69905, the experimental structure was already available, so the appearance of an AlphaFold prediction likely did not add very much additional insight. However, in the case of Q9NPJ8, AlphaFold represents a meaningful increase in information about this protein’s structure. Thus, we treat the previously solved structures as the “control” group, and the previously unsolved structures as the “treatment” group.²⁴

Because some structures in SwissProt are nearly identical, we take this into account when classifying proteins as either solved or unsolved. Proteins in SwissProt are assigned to mutually exclusive clusters based on their amino acid similarity, again using the MMseqs2 algorithm. If a structure in SwissProt has not been previously solved, but a structure that is in the same 100% similarity cluster *has* been previously solved, we also call the first structure previously solved. This way, we are not calling structures unsolved if they have a near-identical neighbor with an experimental structure, since researchers could use that related structure.²⁵

This information shock provided by predicted structures might be useful for scientists working on questions related to these previously unsolved proteins. Using the protein-linked literature section of UniProt, we test whether there was a change in research intensity among these previously unsolved proteins that have new AI-predicted structures, relative to those proteins that already had an experimental structure.

We note that these two groups of proteins are observably different. Table 3 compares pre-2021 characteristics of SwissProt proteins that were solved vs. unsolved and shows clear differences. The first three rows show that previously solved structures have about 10 to 20 times as many papers published about them on average (excluding structure papers) in

²⁴ This empirical strategy is similar in spirit to Hornbeck (2010), though the two papers study entirely different settings. In both contexts, there is near-universal adoption of a new low-cost technology (protein structure prediction; barbed wire fencing), and treatment intensity is determined by access to a pre-existing substitute (experimental protein structures; local woodland endowment that made wooden fencing viable).

²⁵ In practice, have 546,609 unique UniProt IDs and 459,094 unique 100% cluster IDs. In the appendix, we test robustness to different definitions based on broader categories of similarity at the 90%, 50%, and 30% level.

the pre-period depending on the paper count measure. They are more than twice as likely to have at least one paper. Patent activity prior to 2021 is similarly skewed, with solved structures having five to six times as many linked patents, and being more than twice as likely to ever be linked to a patent. This imbalance on pre-period outcomes is not surprising if research clusters on the most biologically relevant proteins.

The stark difference in baseline levels also motivates our choice of a proportional (Poisson pseudo-maximum likelihood, or PPML) specification over a linear model. In a linear difference-in-differences model, parallel trends requires that the two groups would have added similar numbers of papers or patents in the absence of AlphaFold—an implausible assumption when one group begins with over ten times the research activity of the other. Our PPML specification instead requires that the two groups would have grown at similar rates absent the treatment, which is considerably more credible in our setting. We also note that AlphaFold prediction confidence is nearly identical between both sets of proteins. This aligns with the fact that AlphaFold performs well out of sample, as discussed in Section 2. This is important for our research design, because it implies that our results are not being driven by differentially useful predictions.

We compare how publication rates evolve before and after the introduction of AlphaFold differentially for experimentally unsolved and solved proteins. Our regression sample is a panel of 546,609 SwissProt-indexed proteins in years 2017 through 2025. We estimate a PPML difference-in-differences regression for protein i in year t :

$$\mathbb{E}[Y_{it}|Unsolved_i, t] = \exp(\alpha + \lambda Unsolved_i + \sum_{t \neq 2020} \tau_t \cdot D_t + \sum_{t \neq 2020} \kappa_t \cdot D_t \cdot Unsolved_i) \quad (3)$$

where Y_{it} in this case is defined as a count of all non-structure papers. $Unsolved_i$ is an indicator for whether the protein had an experimentally solved structure model in the PDB prior to AlphaFold’s release in July 2021.²⁶ D_t is an indicator for whether the observation occurred in year t . Standard errors are clustered at the 100% protein cluster level.

Among non-structure papers linked to proteins in UniProt, 43% are linked to more than one protein, and 9% are linked to five or more proteins. To handle this non-unique mapping between papers and proteins, we employ three main publication outcome variables: First,

²⁶ One subtlety is that we want to assign proteins to *stable* solved vs. unsolved categories, but a protein’s solved vs. unsolved status evolves over time. For example, a protein that we code as “solved” as of 2021 may have first been solved in 2019. Thus, for part of a pre-period, this “solved” protein was actually unsolved. In practice, this distinction does not matter because of the slow pace of experimental research: fewer than 1.5% of proteins in our sample are solved for the first time between 2017 and 2021. Moreover, the results are virtually identical if we define solved vs. unsolved in 2017, at the start of our sample window.

we count all papers linked to a particular protein, allowing the same paper to be counted multiple times. This is an attractive measure if we think that papers linked to many proteins represent large contributions of knowledge in many parts of the protein space. On the other hand, papers focused on fewer proteins might have more insight per protein. Therefore, our second measure is fractional papers, where we divide each paper by the number of individual proteins it is linked to.²⁷ Lastly, in order to focus on cases with a one-to-one mapping, we count papers linked to exactly one protein.

Results

Figure 10 plots the year-by-treatment interaction coefficients (κ_t 's) with standard errors from Equation 3. Each of the three panels tests a different publication outcome as defined above. First, we notice a relatively consistent pre-trend for each outcome, suggesting that publication activity was evolving at a similar rate before treatment. The standard difference-in-differences assumption of parallel trends assumes that this pattern would have continued through the post-period absent the appearance of AlphaFold. Instead, we find a statistically significant increase in publications about unsolved proteins after AlphaFold, suggesting a shift in attention when AI-predicted structures become available. This marked shift toward research on previously unsolved proteins is apparent across all three publication definitions.

To get a more precise sense of magnitudes, we also report static difference-in-differences estimates in Table 4. The estimates across Columns 1-3 for each outcome suggest that, in the post-period, research on previously unsolved proteins rose by 23 to 32% relative to previously solved proteins.²⁸ The results are statistically significant at the 1% level. These estimates are even larger if we focus on the coefficients from the end of our dynamic specification, suggesting 80 to 300% increases by 2025, depending on the outcome. These results suggest that AlphaFold stimulated significant new basic research about proteins for which we previously lacked an experimental structure model.

We take advantage of the category tags provided for every SwissProt curated paper to analyze how AlphaFold may have impacted a variety of subfields of protein science. We focus on the six largest paper categories: Function (51% of papers are tagged), Expression (13%), Phenotypes (9%), Sequences (8%), Disease (8%), and Interaction (4%).²⁹ Each of these categories represent a major strain of protein research, covering research questions about

²⁷ Note that this outcome will take non-integer values. Our Difference-in-differences estimates follow the Poisson pseudo-maximum likelihood (PPML) approach of Santos Silva and Tenreiro (2006). Consistency of the PPML estimator requires only that the conditional mean is correctly specified; it does not require that outcomes be integer-valued.

²⁸ For example, using the coefficient from Column (2), we compute the percent change by taking $100 \times (e^{0.271} - 1) \approx 31\%$

²⁹ Papers can be tagged with multiple categories, so the same paper can be counted in multiple categories.

how proteins function in cells, how they are expressed from DNA, how they affect disease, how they interact with each other, and more. Figure 11 estimates Equation 3, replacing the fractional publication count with specific fractional counts of papers in each category. We find statistically significant shifts in activity toward previously unsolved proteins in all six paper categories. The largest category—function papers—reacts most quickly. Expression, phenotypes, sequences, disease, and interaction categories increase later, starting around 2023, but the shift is large by 2025. These results suggest that AlphaFold is having broad impacts on a variety of basic research topics in protein science, with multiple scientific specialties benefiting from the shock of new structural insights.

Analysis of AlphaFold citation rates

One potential threat to our identification strategy would be if an unrelated technology or research demand shock occurred around 2021, leading us to misattribute the treatment effect to AlphaFold instead of the other unobserved but correlated shocks. We can directly test the role of AlphaFold in driving this shift toward unsolved protein by tracking citations to core AlphaFold papers. EMBL-EBI strongly encourages any scientists using the AlphaFold model or referencing structure predictions on the AlphaFold DB to cite three relevant papers: Jumper et al., 2021; Varadi, Anyango, et al., 2022; and Varadi, Bertoni, et al., 2024.³⁰ Although these citations are recommended, we recognize that they may be under-provided by authors either for strategic reasons or inattention. Nevertheless, we can compare the citation rates to these three papers after the AlphaFold release separately for papers about solved vs. unsolved proteins.

To implement this, we use the references lists in SwissProt publications linked from OpenAlex.³¹ We then split the papers into a previously solved or unsolved group based on the proteins linked to the paper.³² Figure 12 shows the share of papers that cite at least one of the three AlphaFold papers separately for both groups. Panel (a) shows the raw counts and panel (b) estimates the difference using a similar specification to Equation 2.

We first notice that by 2024, about 2% of all SwissProt-linked papers cite one of the three AlphaFold papers. On one hand, almost no academic papers reach the level of influence so as to be cited by 2% of papers across a massive swath of science.³³ On the other hand, 2% is a relatively modest share given the massive impact AlphaFold is having in this sphere.

³⁰ See <https://www.ebi.ac.uk/training/online/courses/alphafold/accessing-and-predicting-protein-structures-with-alphafold/how-to-cite-alphafold/>

³¹ Our OpenAlex data are from early 2025, thus for this analysis we stop in 2024.

³² If the paper is linked to a mix of previously solved and unsolved proteins, we categorize it as unsolved.

³³ By the end of 2025, Jumper et al., 2021 had more than 46,000 citations on Google Scholar, though many of them from computer scientists and computational biologists.

This might reflect low adherence to the citation norm encouraged by EMBL-EBI.³⁴ Putting aside the base citation rate, the key fact that we highlight is the stark difference in citation rates between papers about solved and unsolved proteins. By 2024, the unsolved group is citing the core AlphaFold papers at almost double the rate of the solved group. This result underscores that AlphaFold is likely the primary driver in the shift of direction in basic research toward previously unsolved proteins.

4.3 Robustness and alternative specifications

Investigating the Covid-19 pandemic

One potential confounder in this analysis is the intense shift in attention toward the SARS-CoV-2 proteins during the Covid-19 pandemic (Hill, Yin, et al., 2025). SwissProt includes 17 proteins that have SARS-CoV-2 listed as their organism. Although these proteins are a tiny fraction of SwissProt, they are linked to about 1,100 individual papers, about half a percent of all papers published about proteins since 2020. DeepMind and EMBL-EBI decided to not post AlphaFold predictions of proteins found in viruses, so we drop these by default from the main analysis. Nonetheless, [Appendix Figure 8](#) shows that including these Covid-relevant papers and proteins does not noticeably impact our results.

Alternative definitions of solved vs. unsolved proteins

We also show robustness to alternative definitions of solved and unsolved in 2021. Our main specification uses a relatively narrow 100% sequence similarity definition to tag proteins that had a structure solution at the time of AlphaFold release. We could instead define a protein as solved as long as a 90% similar protein has been solved, using amino acid sequence similarity. We repeat for different levels of similarity (50% and 30%). Using a broader definition increases the share of proteins that are considered solved. We don't take a strong stand on the correct breadth to define solved, which depends in practice on how exactly scientists are using these structures. In [Appendix Table 3](#), we show that all three publication outcomes are robust to broader definitions of unsolved. The estimated treatment effects are slightly lower for the broader categories (7 to 19% for the 30% similarity definition) but still statistically significant at the 1% level.

³⁴ In a manual inspection of 100 SwissProt-linked papers, we found that fewer than half of papers that mentioned "AlphaFold" included a citation.

4.4 Spillovers to downstream R&D

AlphaFold provides structure predictions that may also be useful in downstream commercial R&D. We test this by again comparing research activity about proteins that had previously been solved experimentally to those that had not. We measure downstream R&D using patent counts, linking proteins to patents with UniProt IDs assigned to patents by SureChEMBL. We count both “all patents” and “medical patents,” restricting to patents with IPCR codes that correspond to medical-related fields (Schmoch, 2008). In both cases, we only count one patent per family to avoid double-counting the same invention, and assign the earliest publication date within the family.

With these outcomes, we estimate the same PPML difference-in-differences specification as described in Equation 3, but replace publication counts with patent counts. Figure 13 reports the results of the dynamic specification. The pre-trends are flat. In the post-period, we see modest increases in unsolved structures relative to solved ones, though these coefficients are not statistically significantly different from zero until 2025. In 2025, we see increases of five to ten percent relative to solved structures that are significant at the five percent level.

Perhaps more importantly, however, our standard errors allow us to rule out large effects. Looking at columns (4) and (5) of Table 4, we see that the relative change in the post-period is between -1% and 13% for all patents, and -2% and 7% for medical patents.³⁵ This lets us conclude the the relative increase in R&D among unsolved proteins is substantially smaller than the relative increase in basic science.

We note that these results are in contrast to the findings of Tranchero (2024), who in a similar difference-in-differences design finds that patenting increases on gene-disease pairs after new information is revealed linking the gene and disease. He observes relative increases beginning in the first year after the information is revealed, and peaking in the fourth to fifth years at just over 0.2 patent applications, an increase of over 100%. His results suggest that it is not simply the case that insufficient time has elapsed for firms to respond. Rather, the lack of response in patenting likely has a different cause.

4.5 Discussion

What should we make of the contrasting results between basic science and commercial R&D? One possibility is that while AlphaFold illuminated a massive number of structures, most of the “useful” proteins had already been experimentally solved. As a collective community,

³⁵ Using column (4) as an example, the bounds are computed by taking $100 \times (e^{0.053 \pm 1.96 \times 0.034} - 1) \approx -1\%$ to 13% .

structural biologists spent five decades slowly tackling the structures of the most disease-relevant proteins. Although they had not fully completed the determination of the entire proteome, they plucked much of the low-hanging fruit that was relevant for many diseases. However, the scientific evidence pushes back against this interpretation. In 2014, the National Institutes of Health launched the Illuminating the Druggable Genome (IDG) Program, explicitly to tackle the fact that pharmaceutical research has clustered on a small set of proteins. The IDG estimates that there are as many as 4,500 proteins susceptible to targeting by drugs, but only about 15% of these are targeted by FDA-approved drugs (Sharma et al., 2024). Much of the attention remains on proteins that were known prior to the mapping of the human genome in the early 2000s, which likely reflects path-dependence rather than scientific potential (Edwards et al., 2011).

An alternative possibility is that protein structure alone is insufficient to initiate the process of drug discovery or other downstream innovation. These experimentally unsolved proteins may remain poorly understood, even after the arrival of predicted structures. As Table 3 shows, these proteins had far fewer papers written about them prior to AlphaFold. Before researchers will invest in developing drugs that target a given protein, that protein’s function and disease relevance must be sufficiently established. Establishing this kind of causal disease relevance requires extensive experimental validation in cell and animal models. Historically, structural research and functional or disease research were symbiotic and often proceeded in parallel: evidence implicating a protein in a disease pathway would motivate structural biologists to solve its structure, while a new structure could in turn generate hypotheses about function and mechanism. AlphaFold represented a massive shock of structural information, but without the complementary biological context, new drug targets will not necessarily emerge. Our evidence suggests that this new influx of structural information *is* leading to more basic science on these proteins, and so this biological context may be on its way. For example, Figure 11 shows that basic science about how proteins relate to disease has markedly increased for these newly illuminated proteins. In the short run, however, it may be that this science is playing catch-up. This may imply that the pharmaceutical R&D is coming in the future, but for now is rate-limited by additional human-conducted science that is currently underway. Of course, we caution that much of this remains speculative, and future research will be needed to see how this plays out in subsequent years.

5 Conclusion

AlphaFold offers a clean, early view of how modern AI is already impacting scientific discovery. In the narrow domain where it most plausibly substitutes for humans—experimental

structure determination—we find little evidence so far of displacement. Experimental deposits in the PDB and publication outcomes remain stable in the four years after AlphaFold’s release, and researchers are continuing to start new experimental work. At the same time, the technology appears to be changing *how* experimentalists work. The rapid expansion of molecular replacement using AlphaFold templates, especially for proteins without close experimental homologs, suggests that humans are using the AI output to make previously challenging experimental work more efficient. Whether or not AI-driven complementarity will persist as prediction tools improve and expand their capabilities remains to be seen.

Downstream of experimental structure determination, we find a different pattern that might be representative of other scientific settings affected by AI. AlphaFold delivered a broad, low-cost knowledge shock to researchers who consume structural information, and we observe a large relative increase in non-structure publications about previously unsolved proteins in the years after release. AlphaFold seems to have acted as a “floodlight,” expanding the set of proteins that scientists are studying.

Yet this apparent broadening of basic research attention does not translate—at least within our current window—into a comparable shift in downstream applied R&D as measured by patent activity. While speculative, this suggests that even when AI dramatically lowers the cost of a key input (here, structural information), downstream progress may remain constrained by complementary tasks that remain in the hands of humans.

In our view, these findings also underscore why the consequences of AI for science remain uncertain. Even as more steps become automated, new constraints will emerge. Thinking further down the drug development pathway, for example, clinical trials remain a formidable bottleneck. While some may point to the possibility of “end-to-end” AI pipelines (Demirer et al., 2026), the timeline and feasibility of these remain uncertain. Thus, in our view, even vast AI-enabled improvements along the basic science to clinical pathway do not guarantee massive gains in drug approval or human health in the short-to-medium run.

References

- Abramson, Josh et al. (2024). “Accurate structure prediction of biomolecular interactions with AlphaFold 3”. *Nature* 630.8016, pp. 493–500.
- Acemoglu, Daron and David Autor (2011). “Skills, Tasks and Technologies: Implications for Employment and Earnings”. *Handbook of Labor Economics*. Ed. by Orley Ashenfelter and David Card. Vol. 4B. Elsevier. Chap. 12, pp. 1043–1171.
- Acemoglu, Daron and Pascual Restrepo (2019). “Automation and new tasks: How technology displaces and reinstates labor”. *Journal of Economic Perspectives* 33.2, pp. 3–30.
- (2022). “Tasks, automation, and the rise in US wage inequality”. *Econometrica* 90.5, pp. 1973–2016.

- Agarwal, Nikhil et al. (2023). “Combining human expertise with artificial intelligence: Experimental evidence from radiology”. *NBER Working Paper*.
- Aghion, Philippe, Benjamin F. Jones, and Charles I. Jones (2019). “Artificial Intelligence and Economic Growth”. *The Economics of Artificial Intelligence: An Agenda*. Ed. by Ajay Agrawal, Joshua Gans, and Avi Goldfarb. University of Chicago Press.
- Agrawal, Ajay, John McHale, and Alexander Oettl (2023). “Superhuman science: How artificial intelligence may impact innovation”. *Journal of Evolutionary Economics* 33.5, pp. 1473–1517.
- (2024). “Artificial Intelligence and Scientific Discovery: A Model of Prioritized Search”. *Research Policy* 53.5, p. 104989.
- Akdel, Mehmet et al. (2022). “A Structural Biology Community Assessment of AlphaFold2 Applications”. *Nature Structural & Molecular Biology* 29.11, pp. 1056–1067.
- Anfinsen, Christian B. (1973). “Principles that Govern the Folding of Protein Chains”. *Science* 181.4096, pp. 223–230.
- Autor, David H. (2015). “Why Are There Still So Many Jobs? The History and Future of Workplace Automation”. *Journal of Economic Perspectives* 29.3, pp. 3–30.
- Autor, David H., Lawrence F. Katz, and Alan B. Krueger (1998). “Computing Inequality: Have Computers Changed the Labor Market?” *The Quarterly Journal of Economics* 113.4, pp. 1169–1213.
- Autor, David H., Frank Levy, and Richard J. Murnane (2003). “The Skill Content of Recent Technological Change: An Empirical Exploration”. *The Quarterly Journal of Economics* 118.4, pp. 1279–1333.
- Ball, Philip (2024). “How AI Revolutionized Protein Science, but Didn’t End It”. *Quanta Magazine*. Accessed 2026-02-16.
- Boustan, Leah Platt, Jiwon Choi, and David Clingingsmith (2024). “Computerized Machine Tools and the Transformation of US Manufacturing”. *NBER Working Paper*.
- Brynjolfsson, Erik, Danielle Li, and Lindsey Raymond (2025). “Generative AI at work”. *The Quarterly Journal of Economics* 140.2, pp. 889–942.
- Callaway, Ewen (2020). “‘It will change everything’: DeepMind’s AI makes gigantic leap in solving protein structures”. *Nature* 588.7837, pp. 203–205.
- (2025). “AlphaFold is five years old — these charts show how it revolutionized science”. *Nature* 648.8093. News article; published 26 Nov 2025 (with correction on 27 Nov 2025)., pp. 258–259.
- Cockburn, Iain M., Rebecca Henderson, and Scott Stern (2019). “The Impact of Artificial Intelligence on Innovation”. *The Economics of Artificial Intelligence: An Agenda*. Ed. by Ajay Agrawal, Joshua Gans, and Avi Goldfarb. University of Chicago Press.
- Cui, Zheyuan Kevin et al. (2025). “The effects of generative AI on high-skilled work: Evidence from three field experiments with software developers”. *Working Paper*.
- Demirer, Mert et al. (2026). “Chaining Tasks, Redefining Work: A Theory of AI Automation”. *NBER Working Paper*.
- Deneke, Victoria E. et al. (2024). “A conserved fertilization complex bridges sperm and egg in vertebrates”. *Cell* 187.25. Epub 2024-10-17., 7066–7078.e22.
- Edich, Maximilian et al. (2022). “The Impact of AlphaFold2 on Experimental Structure Solution”. *Faraday Discussions* 240, pp. 184–195.

- Edmunds, Nicholas S., Liam J. McGuffin, and Ahmet G. Genc (2024). “Benchmarking of AlphaFold2 accuracy self-estimates as indicators of empirical model quality and ranking: a comparison with independent model quality assessment programmes”. *Bioinformatics* 40.8, btae480.
- Edwards, Aled M. et al. (2011). “Too many roads not taken”. *Nature* 470.7333, pp. 163–165.
- Erickson, J. et al. (1990). “Design, Activity, and 2.8 Å Crystal Structure of a C2 Symmetric Inhibitor Complexed to HIV-1 Protease”. *Science* 249.4968, pp. 527–533.
- Evans, Richard et al. (2022). “Protein complex prediction with AlphaFold-Multimer”. *bioRxiv*. Preprint.
- Feigenbaum, James and Daniel P Gross (2024). “Answering the call of automation: How the labor market adjusted to mechanizing telephone operation”. *The Quarterly Journal of Economics* 139.3, pp. 1879–1939.
- Gans, Joshua S. and Myra Mohnen (2026). “How AI Impacts the Quest for Knowledge: Evidence from AlphaFold2”. *Working paper*.
- Goldberg, Samuel and H Tai Lam (2025). “Generative AI in equilibrium: Evidence from a creative goods marketplace”. *Working Paper*.
- Hill, Ryan and Carolyn Stein (2025a). “Race to the Bottom: Competition and Quality in Science”. *Quarterly Journal of Economics* 140.2, pp. 1111–1185.
- (2025b). “Scooped! Estimating Rewards for Priority in Science”. *Journal of Political Economy* 133.3, pp. 793–845.
- Hill, Ryan, Yian Yin, et al. (2025). “The pivot penalty in research”. *Nature* 642.8069, pp. 999–1006.
- Hornbeck, Richard (2010). “Barbed Wire: Property Rights and Agricultural Development”. *Quarterly Journal of Economics* 125.2, pp. 767–810.
- Humlum, Anders (2021). “Robot Adoption and Labor Market Dynamics”. *Working Paper*.
- Humlum, Anders and Emilie Vestergaard (2026). “Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI”. *Working Paper*.
- International Human Genome Sequencing Consortium (2004). “Finishing the Euchromatic Sequence of the Human Genome”. *Nature* 431.7011, pp. 931–945.
- Jones, Benjamin (2025). “Artificial intelligence in research and development”. *NBER Working Paper*.
- Jones, Charles I. (1995). “R&D-Based Models of Economic Growth”. *Journal of Political Economy* 103.4, pp. 759–784.
- (2005). “Growth and Ideas”. *Handbook of Economic Growth*. Ed. by Philippe Aghion and Steven N. Durlauf. Vol. 1. Elsevier. Chap. 16, pp. 1063–1111.
- Jumper, John et al. (2021). “Highly accurate protein structure prediction with AlphaFold”. *Nature* 596.7873, pp. 583–589.
- Kim, Rachel Seongeun et al. (2025). “BFVD—a large repository of predicted viral protein structures”. *Nucleic acids research* 53.D1, pp. D340–D347.
- Kim, Soomi (2026). “Navigating the Rugged Data Landscape: The Impact of Data-Extrapolation Technologies on Knowledge Production”. *Working Paper*.
- Kovalevskiy, Oleg, Juan Mateos-Garcia, and Kathryn Tunyasuvunakool (2024). “AlphaFold two years on: Validation and impact”. *Proceedings of the National Academy of Sciences* 121.34, e2315002121.

- Kryshtafovych, Andriy, John Moult, et al. (2021). “Computational Models in the Service of X-ray and Cryo-electron Microscopy Structure Determination”. *Proteins: Structure, Function, and Bioinformatics* 89.12, pp. 1633–1646.
- Kryshtafovych, Andriy, Torsten Schwede, et al. (2021). “Critical Assessment of Methods of Protein Structure Prediction (CASP)-Round XIV”. *Proteins: Structure, Function, and Bioinformatics* 89.12, pp. 1607–1617.
- Levinthal, Cyrus (1969). “How to Fold Graciously”. *Mössbauer Spectroscopy in Biological Systems*. Ed. by P. Debrunner, J. C. M. Tsibris, and E. Münck. Urbana, IL: University of Illinois Press, pp. 22–24.
- Ludwig, Jens and Sendhil Mullainathan (2024). “Machine Learning as a Tool for Hypothesis Generation”. *The Quarterly Journal of Economics* 139.2, pp. 751–827.
- McCoy, Airlie J. et al. (2007). “Phaser Crystallographic Software”. *Journal of Applied Crystallography* 40.4, pp. 658–674.
- Metzker, Michael L. (2010). “Sequencing Technologies — The Next Generation”. *Nature Reviews Genetics* 11.1, pp. 31–46.
- Moult, John et al. (1995). “A large-scale experiment to assess protein structure prediction methods”. *Proteins: Structure, Function, and Genetics* 23.3, pp. ii–v.
- Mullainathan, Sendhil and Ashesh Rambachan (2025). “Science in the Age of Algorithms”. *The Economics of Transformative AI*. NBER.
- Nagaraj, Abhishek (2021). “The Private Impact of Public Data: Landsat Satellite Maps Increased Gold Discoveries and Encouraged Entry”. *Management Science* 68.1, pp. 564–582.
- Nagaraj, Abhishek, Esther Shears, and Mathijs de Vaan (2020). “Improving Data Access Democratizes and Diversifies Science”. *Proceedings of the National Academy of Sciences* 117.38, pp. 23490–23498.
- Navia, Manuel A et al. (1989). “Three-dimensional structure of aspartyl protease from human immunodeficiency virus HIV-1”. *Nature* 337.6208, pp. 615–620.
- Nogales, Eva (2016). “The Development of Cryo-EM into a Mainstream Structural Biology Technique”. *Nature Methods* 13.1, pp. 24–27.
- Noy, Shakked and Whitney Zhang (2023). “Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence”. *Science* 381.6654, pp. 187–192.
- Papadatos, George et al. (2016). “SureChEMBL: a large-scale, chemically annotated patent document database”. *Nucleic Acids Research* 44.D1, pp. D1220–D1228.
- Phenix (n.d.). *Molecular Replacement: Overview*. https://phenix-online.org/documentation/reference/mr_overview.html. Phenix documentation page. Accessed 2026-02-21.
- Poux, Sylvain et al. (2017). “On expert curation and scalability: UniProtKB/Swiss-Prot as a case study”. *Bioinformatics* 33.21, pp. 3454–3460.
- Priem, Jason, Heather Piwowar, and Richard Orr (2022). “OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts”. *arXiv*. arXiv: [2205.01833](https://arxiv.org/abs/2205.01833) [cs.DL].
- Qian, Jerry (2026). “AI in the Lab: AlphaFold2’s Impacts on Human-Produced Knowledge”. *Working Paper*.
- Restrepo, Pascual (2024). “Automation: Theory, evidence, and outlook”. *Annual review of economics* 16.1, pp. 1–25.

- Romer, Paul M. (1990). “Endogenous Technological Change”. *Journal of Political Economy* 98.5, Part 2, S71–S102.
- Santos, Rita et al. (2017). “A comprehensive map of molecular drug targets”. *Nature Reviews Drug Discovery* 16.1, pp. 19–34.
- Santos Silva, J. M. C. and Silvana Tenreyro (2006). “The Log of Gravity”. *The Review of Economics and Statistics* 88.4, pp. 641–658.
- Schmoch, Ulrich (2008). *Concept of a Technology Classification for Country Comparisons*. Final Report to the World Intellectual Property Organisation (WIPO). Prepared by Fraunhofer Institute for Systems and Innovation Research (ISI). Karlsruhe: World Intellectual Property Organization (WIPO).
- Sharma, Karlie R et al. (2024). “Illuminating the druggable genome: Pathways to progress”. *Drug Discovery Today* 29.3, p. 103805.
- Steinegger, Martin and Johannes Söding (2017). “MMseqs2 Enables Sensitive Protein Sequence Searching for the Analysis of Massive Data Sets”. *Nature Biotechnology* 35.11, pp. 1026–1028.
- Strasser, Bruno J. (2019). *Collecting Experiments: Making Big Data Biology*. Chicago: University of Chicago Press.
- Sullivan, Kevin P., Peggy Brennan-Tonetta, and Lori J. Marxen (2017). “Economic Impacts of the Research Collaboratory for Structural Bioinformatics (RCSB) Protein Data Bank”. *RCSB Protein Data Bank* 10.
- Sun, Mengyi, Sukwoong Choi, and Yian Yin (2026). “AI Predictions and the Expansion of Scientific Frontiers: Evidence from Structural Biology”. *Working paper*.
- Terwilliger, Thomas C. et al. (2024). “AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination”. *Nature Methods* 21.1, pp. 110–116.
- The UniProt Consortium (2025). “UniProt: the Universal Protein Knowledgebase in 2025”. *Nucleic Acids Research* 53, pp. D609–D617.
- Tranchoero, Matteo (2024). “Finding Diamonds in the Rough: Data-Driven Opportunities and Pharmaceutical Innovation”. *Working Paper*.
- Tunyasuvunakool, Kathryn et al. (2021). “Highly accurate protein structure prediction for the human proteome”. *Nature* 596.7873, pp. 590–596.
- Varadi, Mihaly, Stephen Anyango, et al. (2022). “AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models”. *Nucleic Acids Research* 50.D1, pp. D439–D444.
- Varadi, Mihaly, Damian Bertoni, et al. (2024). “AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences”. *Nucleic Acids Research* 52.D1, pp. D368–D375.
- Varadi, Mihaly and Sameer Velankar (2023). “The impact of AlphaFold Protein Structure Database on the fields of life sciences”. *Proteomics* 23.17, p. 2200128.
- Weitzman, Martin L. (1998). “Recombinant Growth”. *The Quarterly Journal of Economics* 113.2, pp. 331–360.
- Wlodawer, Alexander et al. (1989). “Conserved folding in retroviral proteases: crystal structure of a synthetic HIV-1 protease”. *Science* 245.4918, pp. 616–621.
- Wrapp, Daniel et al. (2020). “Cryo-EM Structure of the 2019-nCoV Spike in the Prefusion Conformation”. *Science* 367.6483, pp. 1260–1263.

- Yu, Zhengyi (2026). “The Impacts of AI at Scale: Evidence from Research Scientists”. *Working Paper*.
- Zemla, Adam (2003). “LGA: a method for finding 3D similarities in protein structures”. *Nucleic Acids Research* 31.13, pp. 3370–3374.
- Zhou, Eric and Dokyun Lee (2024). “Generative artificial intelligence, human creativity, and art”. *PNAS Nexus* 3.3, p. 052.
- Zhuo, Ran (2025). “Exploit or Explore? An Empirical Study of Resource Allocation in Scientific Labs”. *Working Paper*.

Tables and Figures

Table 1: Summary statistics: PDB structures

	Mean	Median	SD	Min	Max	Obs
<i>Panel A. Variables relevant to all structures</i>						
Entity count	2.80	1.00	7.57	1.00	165.00	71,133
Residue count	1833	672	8548	3	761922	71,133
Similarity to nearest neighbor	75.0	95.1	33.0	0.0	100.0	71,133
Share that have experimental homolog	0.87	1.00	0.33	0.00	1.00	71,133
AlphaFold prediction confidence	85.9	88.9	10.4	27.0	98.8	59,403
Experimental determination method						
X-ray crystallography	0.72	1.00	0.45	0.00	1.00	71,133
Cryo electron microscopy	0.24	0.00	0.43	0.00	1.00	71,133
Other	0.04	0.00	0.19	0.00	1.00	71,133
<i>Panel B. Variables relevant to x-ray structures</i>						
Use molecular replacement (MR)	0.88	1.00	0.33	0.00	1.00	51,545
List a starting model	0.84	1.00	0.37	0.00	1.00	51,545
List an AlphaFold starting model	0.04	0.00	0.20	0.00	1.00	51,545

Notes: This table presents summary statistics for the experimental PDB structures in our analysis sample over the 2017 to Q1 2025 time frame. Panel A presents statistics relevant for all 71,133 structures. Panel B presents statistics only relevant for the 51,545 structures solved using x-ray crystallography. The number of observations for AlphaFold prediction confidence is lower because some PDB structures do not have an AlphaFold counterpart.

Table 2: Summary statistics: SwissProt proteins

	Mean	Median	SD	Min	Max	Obs
Unsolved prior to AlphaFold	0.920	1.00	0.27	0.00	1.00	546,609
SwissProt non-structure papers (all paper count)	3.63	0.00	38.55	0.00	10461	546,609
SwissProt non-structure papers (fractional papers)	1.02	0.00	19.22	0.00	6661	546,609
SwissProt non-structure papers (single-protein papers)	0.55	0.00	12.25	0.00	4333	546,609
Any SwissProt non-structure paper	0.17	0.00	0.38	0.00	1.00	546,609
Patent count (all patents)	9.90	0.00	493.91	0.00	311175	546,609
Patent count (medical patents)	6.01	0.00	232.04	0.00	82235	546,609
Any patent (all patents)	0.17	0.00	0.38	0.00	1.00	546,609
Any patent (medical patents)	0.17	0.00	0.37	0.00	1.00	546,609
Average pLDDT confidence score	87.51	91.21	10.63	25.97	98.75	546,609

Notes: This table presents summary statistics for the proteins indexed by SwissProt. All count variables are counts prior to AlphaFold’s release in 2021. Paper counts come from papers that are linked to proteins by SwissProt and exclude papers that are tagged as structural papers. For fractional papers, if a paper is linked to N proteins, each protein is assigned $1/N$ of the paper. Patent counts come from patents that are linked to proteins by SureChEMBL. For patents, we collapse down to the family level and assign the earliest publication year. Medical patents are patents assigned a IPCR code corresponding to the pharmaceutical, medical technology, or biotechnology fields (Schmoch, 2008). $N = 546,609$ SwissProt-indexed proteins.

Table 3: Summary statistics: SwissProt proteins, solved vs. unsolved structures

	Solved structures	Unsolved structures
SwissProt non-structure papers (all paper count)	20.12	2.19
SwissProt non-structure papers (fractional papers)	7.50	0.46
SwissProt non-structure papers (single-protein papers)	4.37	0.22
Any SwissProt non-structure paper	0.35	0.15
Patent count (all patents)	38.74	7.38
Patent count (medical patents)	26.36	4.23
Any patent (all patents)	0.35	0.16
Any patent (medical patents)	0.34	0.15
Average pLDDT confidence score	87.75	87.49
Observations	43,992	502,617

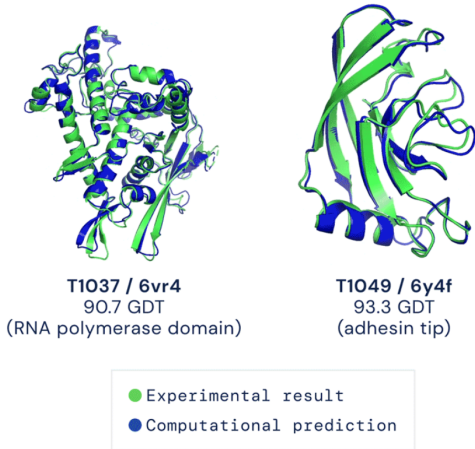
Notes: This table presents means for the proteins indexed by SwissProt, split by whether they have a solved experimental protein structure or not prior to 2021. All variables are counts accrued prior to AlphaFold’s release in 2021. Paper counts come from papers that are linked to proteins by SwissProt and exclude papers that are tagged as structural papers. For fractional papers, if a paper is linked to N proteins, each protein is assigned $1/N$ of the paper. Patent counts come from patents that are linked to proteins by SureChEMBL. For patents, we collapse down to the family level and assign the earliest publication year. Medical patents are patents assigned a IPCR code corresponding to the pharmaceutical, medical technology, or biotechnology fields (Schmoch, 2008). $N = 546,609$ SwissProt-indexed proteins.

Table 4: Difference-in-differences: Papers in related fields and patents

	(1)	(2)	(3)	(4)	(5)
Dependent Variable:	Non-structure papers (all paper count)	Non-structure papers (fractional papers)	Non-structure papers (single protein papers)	Patents (all)	Patents (medical)
Post	-0.703*** (0.0118)	-0.735*** (0.0131)	-0.728*** (0.0138)	0.145*** (0.0248)	0.012 (0.0195)
Unsolved	-2.248*** (0.0322)	-2.700*** (0.0383)	-2.840*** (0.0408)	-1.668*** (0.1266)	-1.863*** (0.1223)
Post x Unsolved	0.205*** (0.0136)	0.271*** (0.0165)	0.277*** (0.0181)	0.053 (0.0340)	0.026 (0.0229)
Observations	4,919,481	4,919,481	4,919,481	4,919,481	4,919,481

Notes: This table presents Poisson difference-in-differences regression estimates comparing previously solved and unsolved proteins before and after AlphaFold. Post is defined as all papers published after 2021. Solved structures are defined based on whether the protein had an experimental structure deposited in the PDB prior to AlphaFold’s release on July 22, 2021. Column 1 outcome is counts of all non-structure papers linked to the protein by SwissProt, including those linked to multiple proteins. Column 2 outcome is fractional papers, defined as one divided by the number of distinct proteins linked to the paper. Column 3 outcome is counts of non-structure papers that map uniquely to a single protein. Column 4 outcome is the count of unique patent families linked to the protein by SureChEMBL. Column 5 outcome is the count of unique medical patent families linked to the protein, where medical is defined as patents assigned a IPCR code corresponding to the pharmaceutical, medical technology, or biotechnology fields (Schmoch, 2008). Standard errors are clustered at the protein level. $N = 4,919,481$ SwissProt protein-years, coming from 546,609 SwissProt proteins. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$

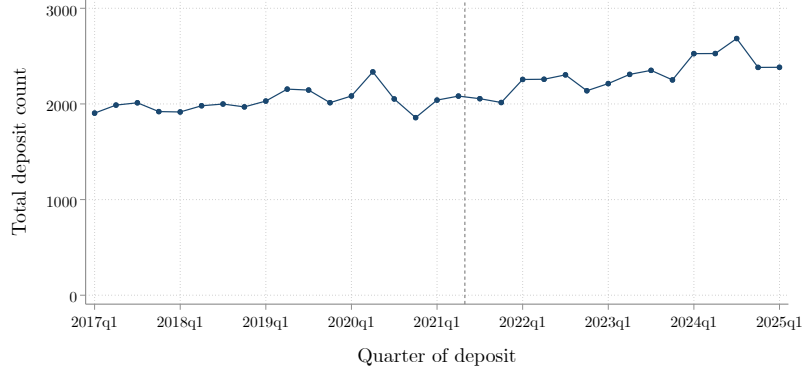
Figure 1: Example of experimental vs. predicted proteins



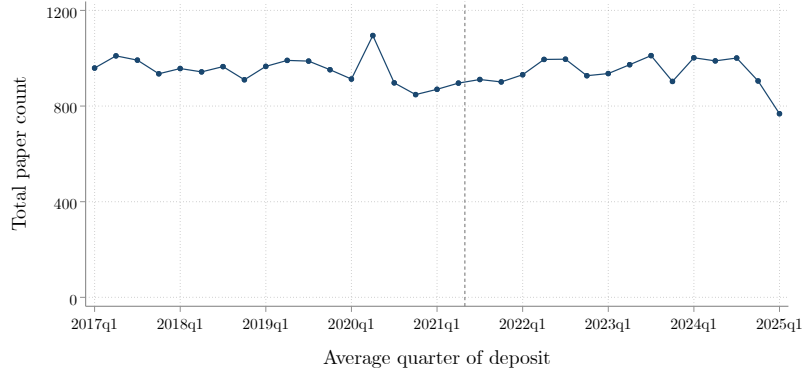
Notes: This figure shows two examples of proteins with their experimental structure overlaid on their predicted structure. Source: DeepMind.

Figure 2: Structural biology experimental output

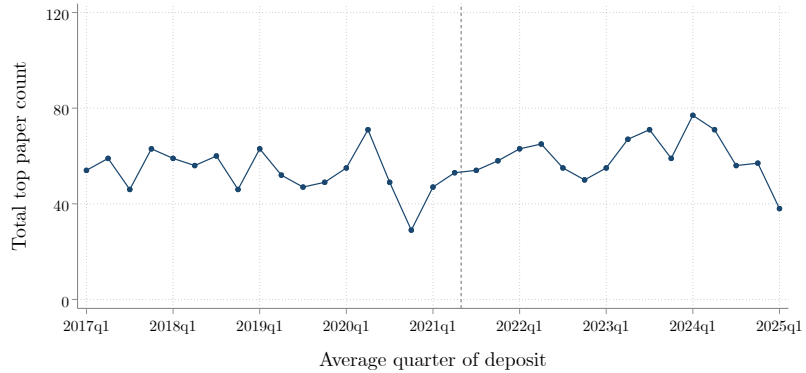
(a) Total PDB deposits



(b) Total paper counts

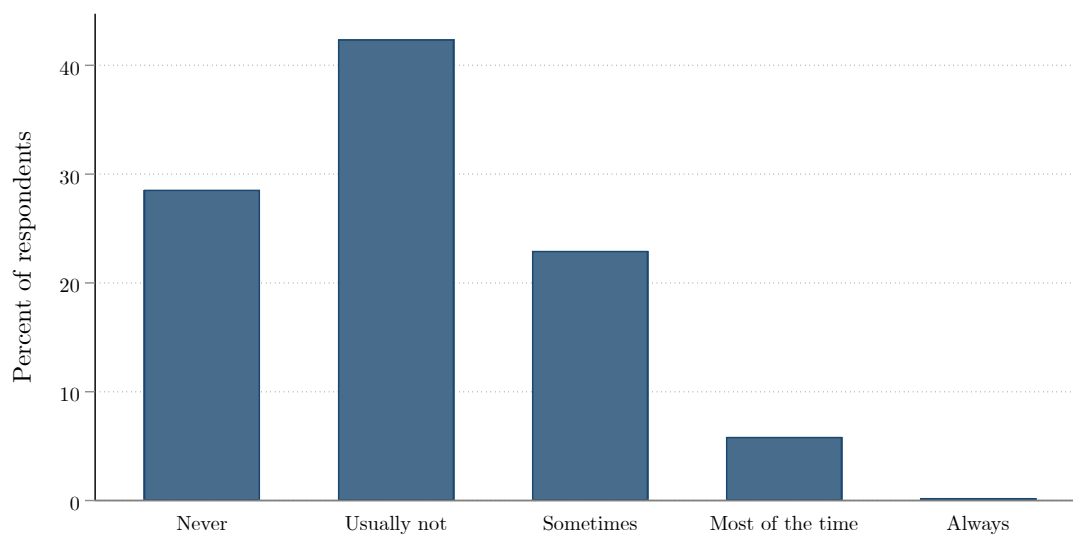


(c) Top paper counts



Notes: This figure shows the total counts of PDB structures and papers in our analysis sample over time. Panel (a) plots PDB deposits based on their deposit dates. Panel (b) collapses structure(s) to the paper level. If there are multiple structures per paper, the paper is assigned the average deposit date of the structures. Panel (c) restricts to “top papers,” defined as papers published in *Cell*, *Nature*, and *Science*. $N = 71,133$ proteins, linked to 31,236 papers.

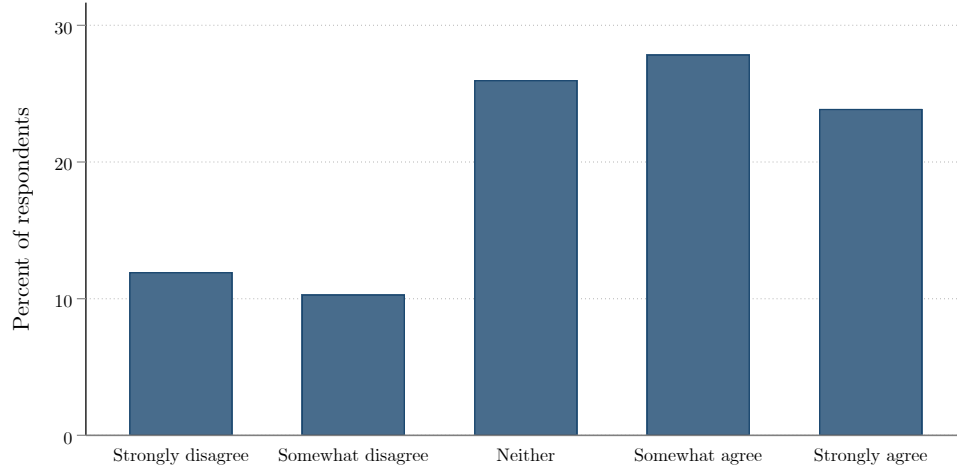
Figure 3: Experimentalist trust in predicted protein structures



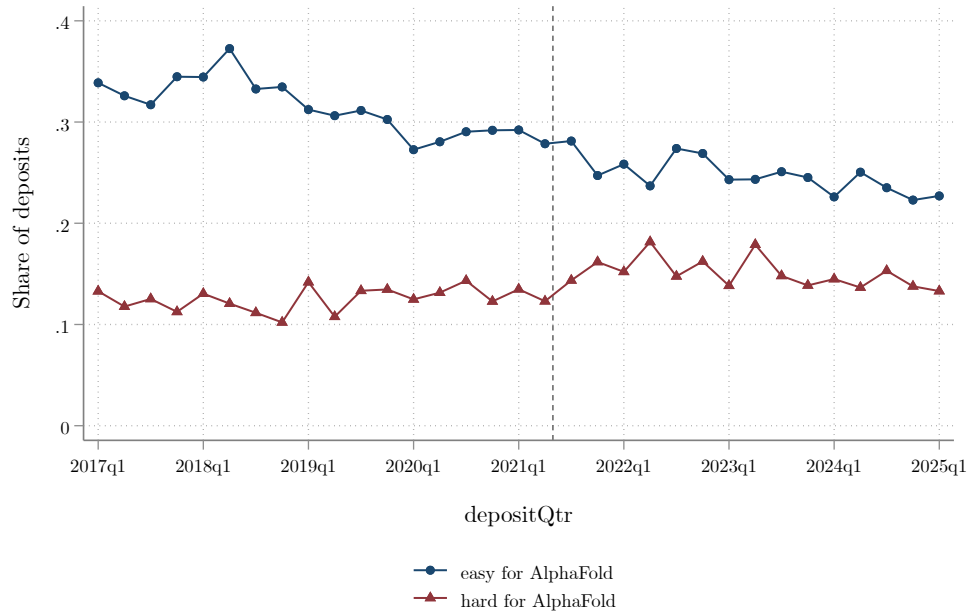
Notes: This figure shows the results of a survey question asking “Do you trust AlphaFold2 or other protein structure predictions enough to skip experimental validation?” $N = 427$ structural biologist survey respondents.

Figure 4: Shifting to experimentalist comparative advantage

(a) Survey: Shift to structures poorly predicted by AlphaFold?



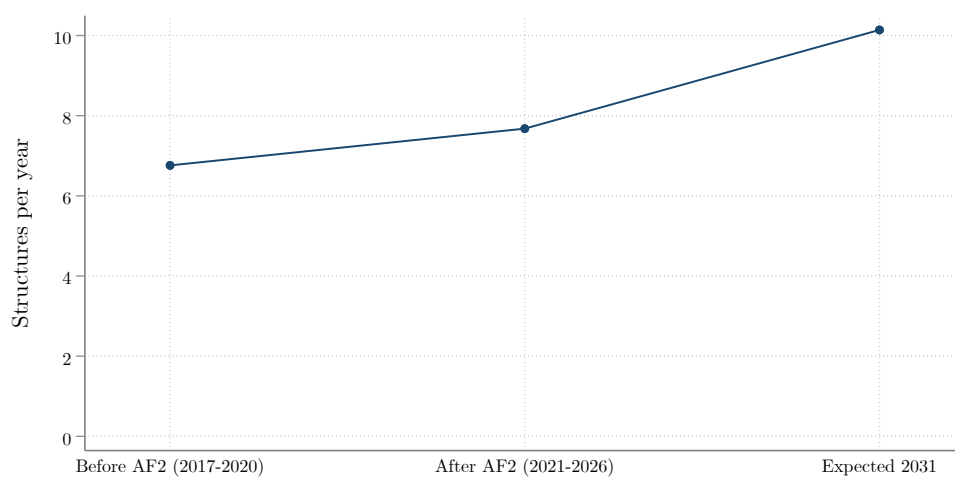
(b) PDB: Share of structures that are easy vs. hard for AlphaFold



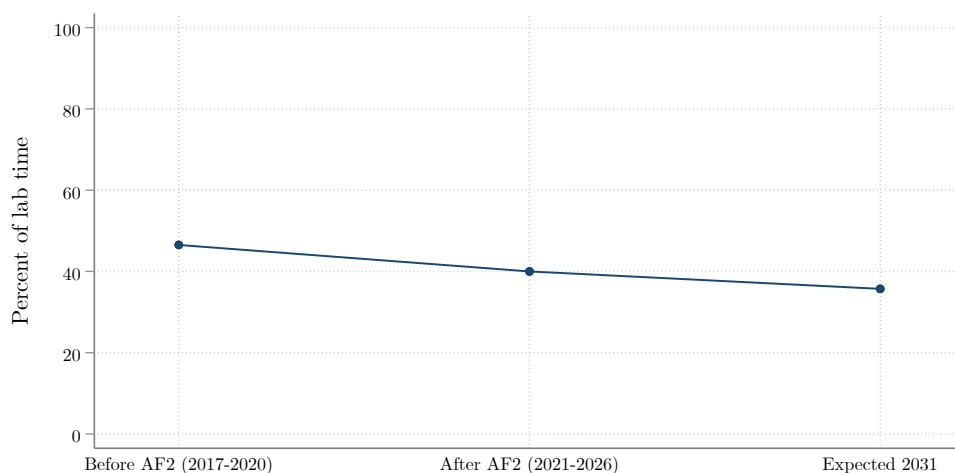
Notes: Panel A shows the results of a survey question asking “Agree or disagree: AlphaFold2 and related advances in protein structure prediction have caused me to shift my research focus toward experimentally solving proteins that are predicted poorly by AI methods.” $N = 427$ structural biologist survey respondents. Panel B shows the share of structures that are coded as being easy or hard for AlphaFold2 to predict. See Appendix B for a discussion of how proteins are classified. $N = 71,133$ proteins.

Figure 5: Experimental productivity over time

(a) Number of experimental structures

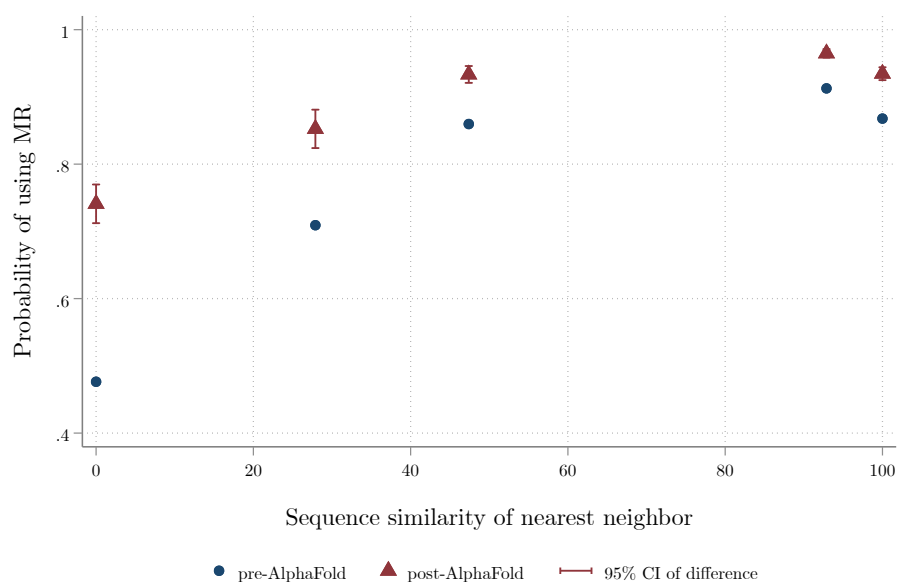


(b) Share of lab's time devoted to experimental structure determination



Notes: Panel A shows the results of a survey question asking “How many experimental protein structures does your lab solve per year?” Panel B shows the results of a survey question asking “What percent of your lab’s total research time is devoted to experimental structure determination?” All respondents answered each question for all three time periods. Each dot shows the mean of all respondents in each period. $N = 427$ structural biologist survey respondents.

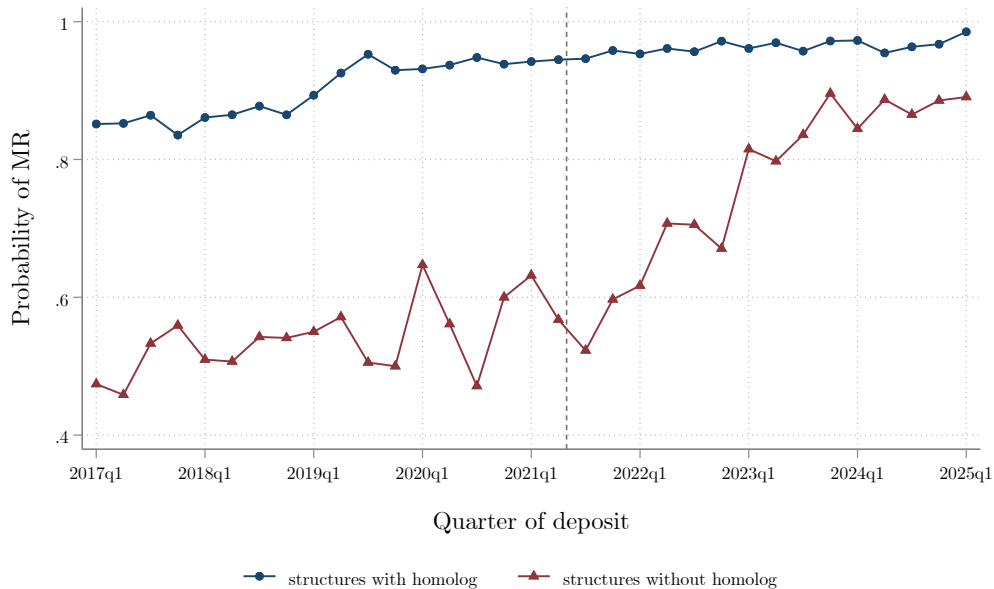
Figure 6: Molecular replacement usage, before and after AlphaFold



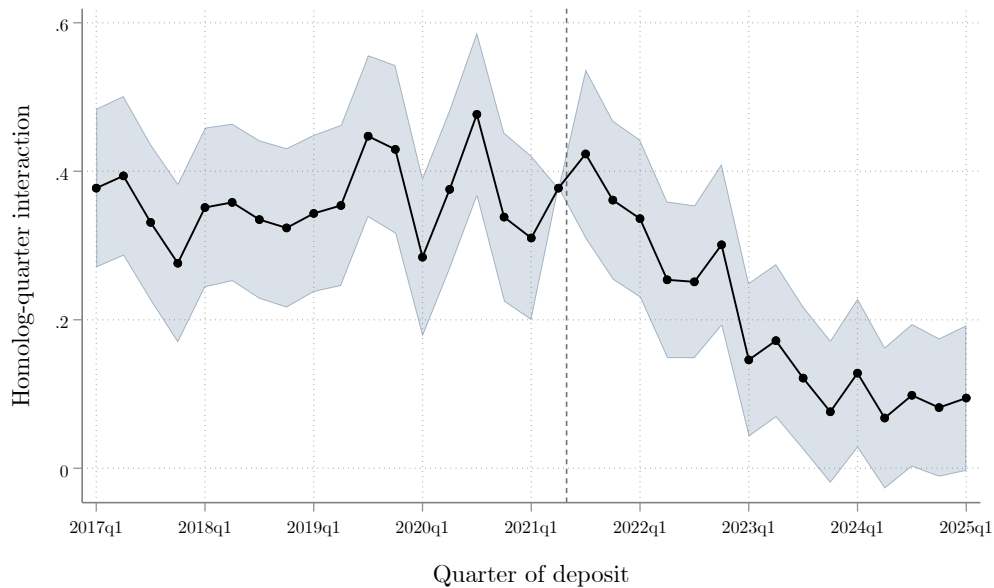
Notes: This figure shows how likely a structure was to be solved by molecular replacement, before and after AlphaFold, following Equation 1 in the text. The x-axis shows the sequence similarity of a protein’s nearest experimentally solved neighbor. The blue series shows the probability of using molecular replacement for structures deposited before AlphaFold, binned by the midpoint of each similarity group (β_g). The red series shows the probability after AlphaFold ($\beta_g + \gamma_g$). The bars show the 95% confidence interval of the difference (γ_g). $N = 51,545$ proteins solved via x-ray crystallography.

Figure 7: Molecular replacement usage over time

(a) Molecular replacement, with versus without homolog



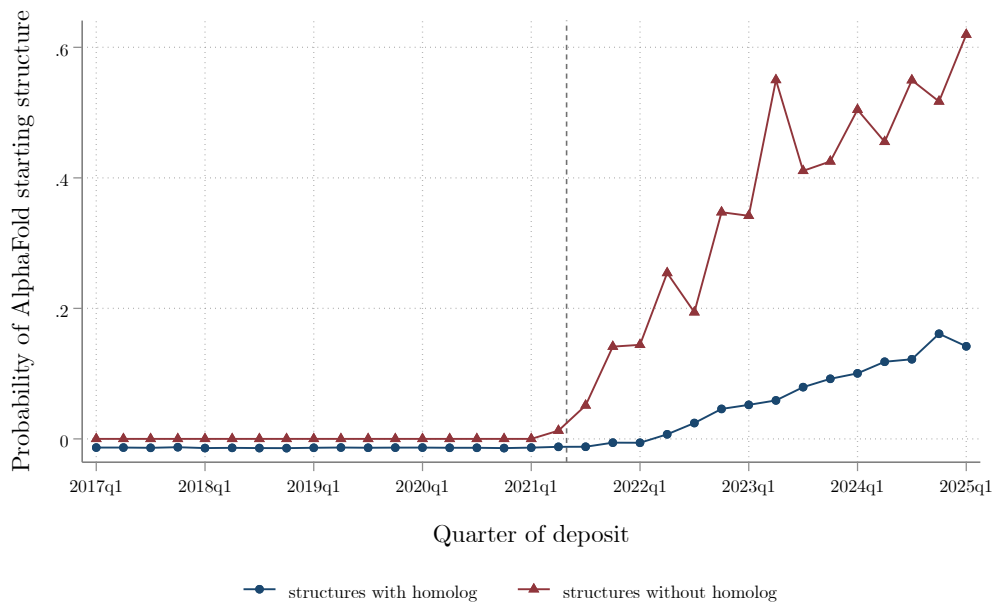
(b) Difference



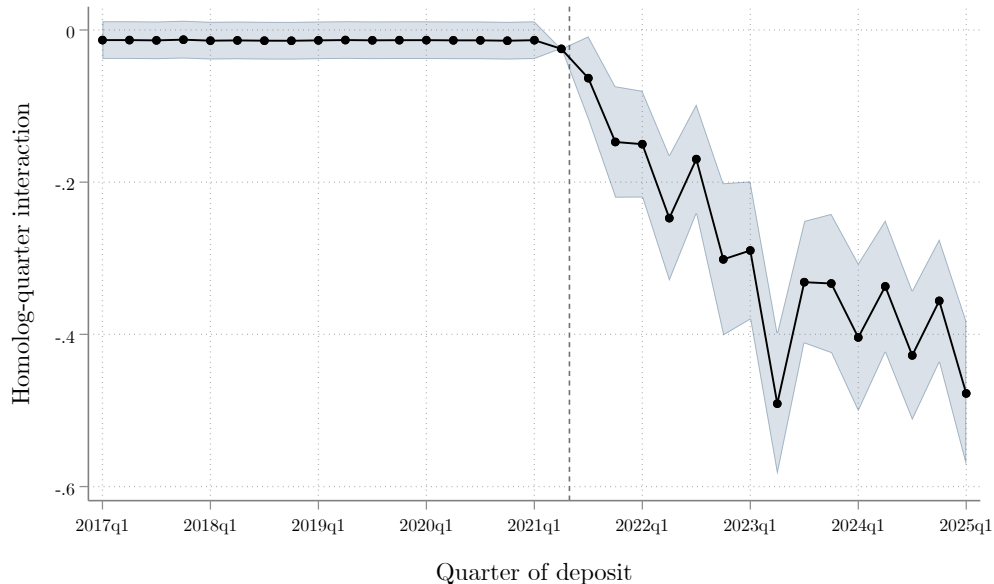
Notes: This figure shows how use of molecular replacement evolved over time, comparing proteins that did and did not have a homolog, following Equation 2 in the text. A protein with a homolog is any protein that had a released protein that was at least 30% similar in terms of amino acid sequence at the time it was deposited. Panel (a) shows the difference in average molecular replacement rates by group, plotting $(\alpha + \delta_q)$ vs $(\alpha + \lambda + \delta_q + \theta_q)$. Panel (b) shows the difference $(\lambda + \theta_q)$, with the shaded area representing the 95% confidence interval. $N = 51,545$ proteins solved via x-ray crystallography.

Figure 8: AlphaFold usage over time

(a) AlphaFold usage, with versus without homolog

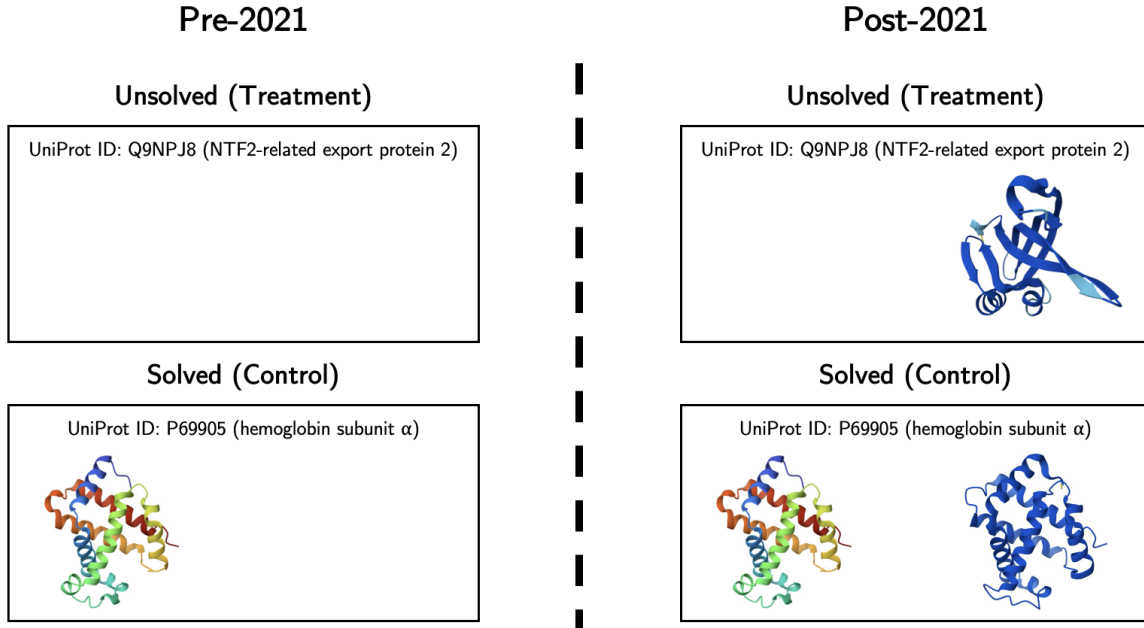


(b) Difference



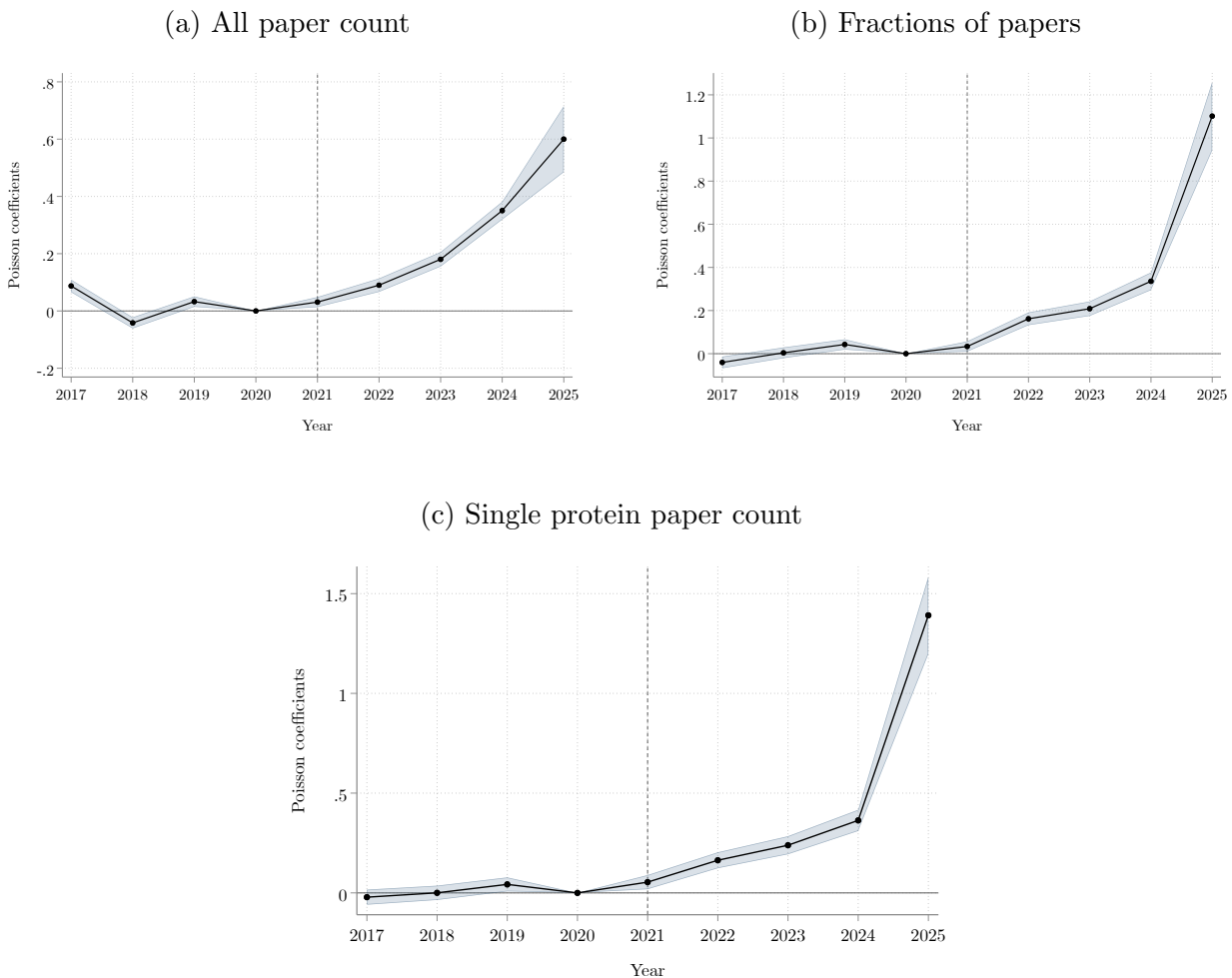
Notes: This figure shows how use of AlphaFold predicted structures as starting structures for molecular replacement evolved over time, comparing proteins that did and did not have a homolog, following Equation 2 in the text. A protein with a homolog is any protein that had a released protein that was at least 30% similar in terms of amino acid sequence at the time it was deposited. Panel (a) shows the difference in average AlphaFold usage by group, plotting $(\alpha + \delta_q)$ vs $(\alpha + \lambda + \delta_q + \theta_q)$. Panel (b) shows the difference $(\lambda + \theta_q)$, with the shaded area representing the 95% confidence interval. $N = 42,007$ proteins solved via x-ray crystallography, using molecular replacement, and listing a starting model.

Figure 9: Difference-in-differences strategy



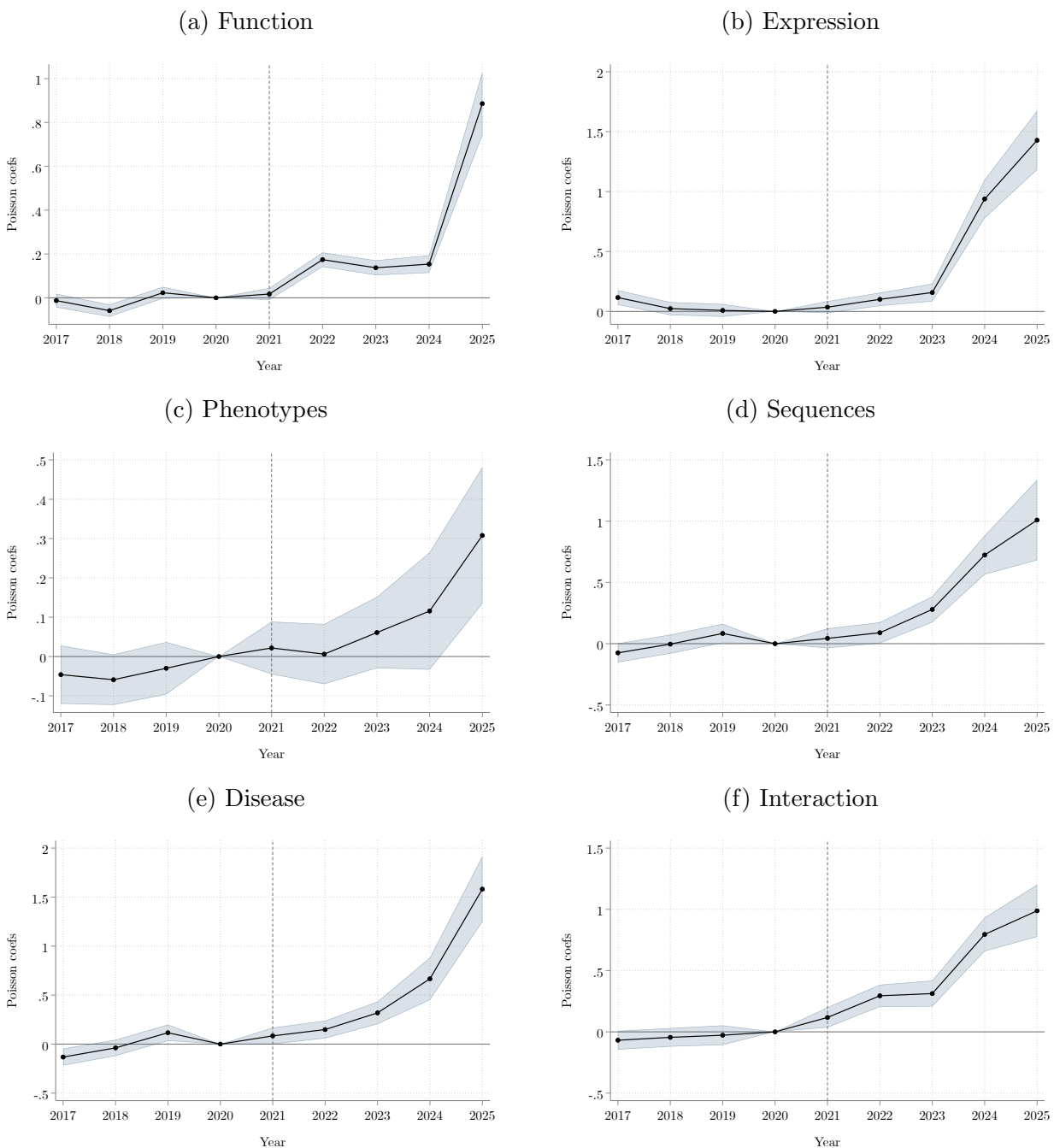
Notes: This figure explains our difference-in-differences strategy. Prior to 2021, UniProt ID Q9NPJ8 had not been solved, so researchers had no structural information about it. Meanwhile, UniProt ID P69905 had been solved and researchers had access to an experimental structure. Post 2021, both structures had a predicted structure. For UniProt ID Q9NPJ8, this represented a large shock to our knowledge of its structure. For UniProt ID P69905, this represented a much smaller knowledge shock, given that we already had the experimental structure. Experimental structure taken from PDB ID 1A01. Predicted structures taken from AlphaFold IDs AF-Q9NPJ8-4-F1 and AF-P69905-F1.

Figure 10: Non-structure publications: solved vs. unsolved proteins



Notes: This figure reports the Poisson difference-in-differences coefficients comparing non-structure paper counts for solved vs. unsolved proteins before and after AlphaFold, following Equation 3 in the text. Solved structures are defined based on whether the protein had an experimental structure deposited in the PDB prior to AlphaFold’s release on July 22, 2021. Panel (a) outcome is counts of all non-structure papers linked to the protein, including those linked to multiple proteins. Panel (b) outcome is fractional papers, defined as one divided by the number of distinct proteins linked to the non-structure paper. Panel (c) outcome is counts of non-structure papers that map uniquely to a single protein. Confidence intervals are shaded in blue and are calculated with standard errors clustered at the protein level. $N = 4,919,481$ SwissProt protein-years, coming from 546,609 SwissProt proteins.

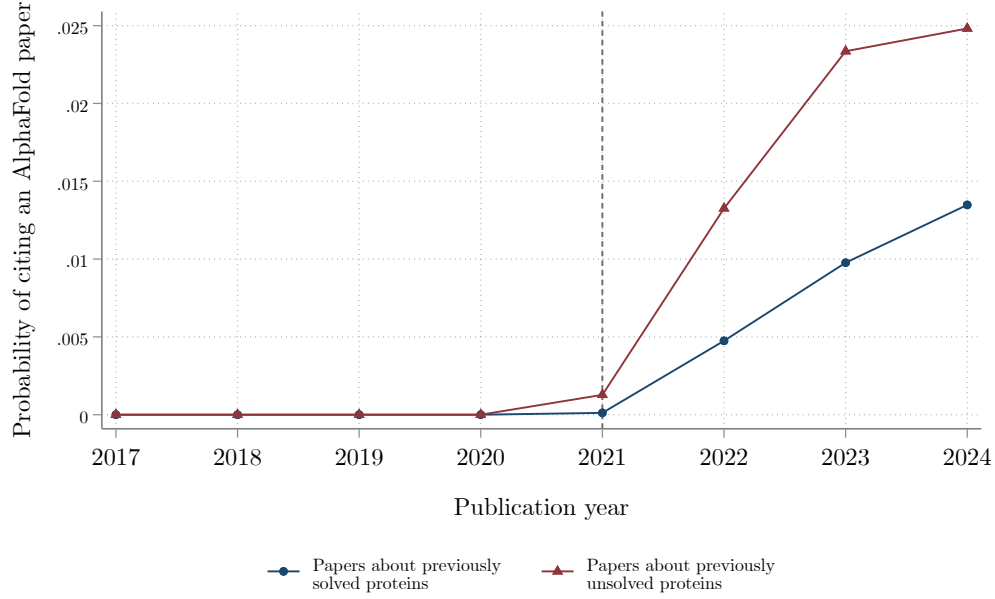
Figure 11: Categories of publications: solved vs. unsolved proteins



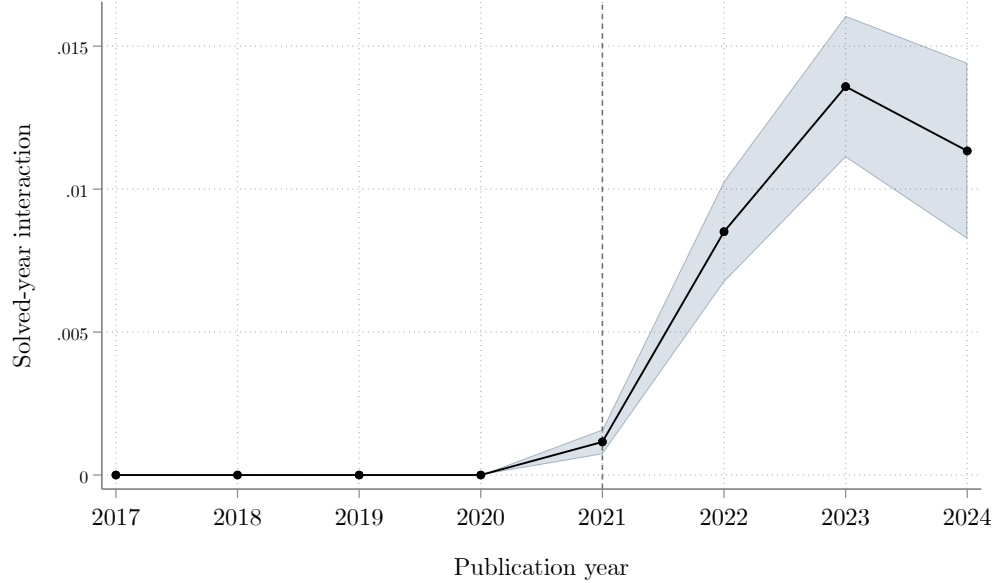
Notes: This figure reports the Poisson difference-in-differences coefficients comparing fractional paper counts for solved vs. unsolved proteins before and after AlphaFold, following Equation 3 in the text. Solved structures are defined based on whether the protein had an experimental structure deposited in the PDB prior to AlphaFold's release on July 22, 2021. Each panel shows a different paper category as assigned in the UniProt literature curation process based on the main basic research topics covered in the paper. Papers can be tagged with multiple categories, and will be counted separately for each category they are tagged with. Confidence intervals are shaded in blue and are calculated with standard errors clustered at the protein level. $N = 4,919,481$ SwissProt protein-years, coming from 546,609 SwissProt proteins.

Figure 12: Citations to AlphaFold papers

(a) Citation share by year, solved vs. unsolved proteins

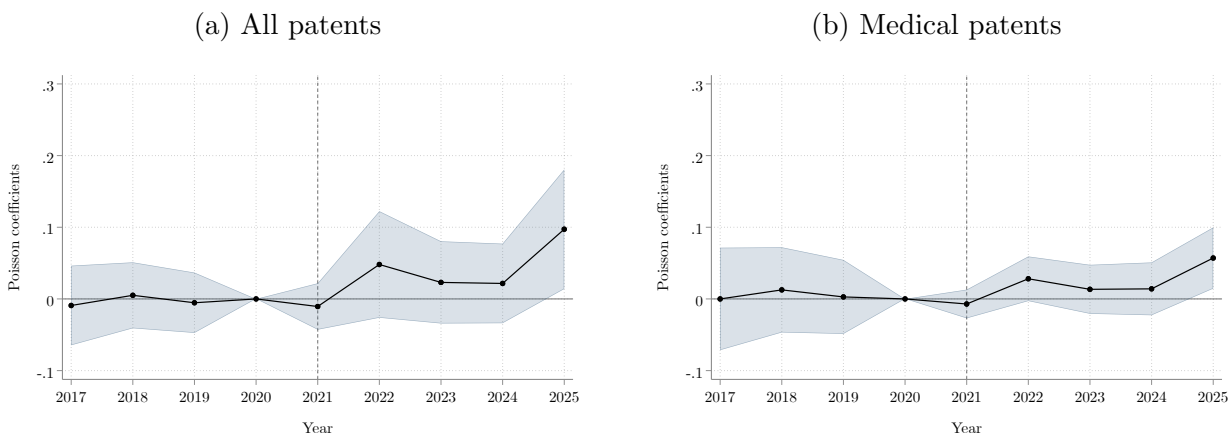


(b) Difference



Notes: This figure shows how citations to AlphaFold papers evolved over time, comparing papers about proteins that were previously solved vs. unsolved. If the paper is linked to any unsolved proteins, we categorize it as unsolved. We define AlphaFold citations as those to any of the three papers EMBL requests authors to cite when using the AlphaFold models or database: Jumper et al., 2021; Varadi, Anyango, et al., 2022; and Varadi, Bertoni, et al., 2024. The regression specification is a version of Equation 2 in the text, adapted to the paper sample. Panel (a) shows the difference in AlphaFold citation rate by group, plotting $(\alpha + \delta_q)$ vs $(\alpha + \lambda + \delta_q + \theta_q)$. Panel (b) shows the difference $(\lambda + \theta_q)$, with the shaded area representing the 95% confidence interval. $N = 382,484$ papers indexed by UniProt from 2017-2024.

Figure 13: Patenting activity: solved vs. unsolved proteins



Notes: This figure reports the Poisson difference-in-differences coefficients comparing SureChEMBL-linked patent counts for solved vs. unsolved proteins before and after AlphaFold, following Equation 3 in the text. Patents are collapsed to the family level, and assigned the earliest publication date within the family. Medical patents are defined as patents assigned a IPCR code corresponding to the pharmaceutical, medical technology, or biotechnology fields (Schmoch, 2008). Solved structures are defined based on whether the protein had an experimental structure deposited in the PDB prior to AlphaFold’s release on July 22, 2021. Confidence intervals are shaded in blue and are calculated with standard errors clustered at the protein level. $N = 4,919,481$ SwissProt protein-years, coming from 546,609 SwissProt proteins.

Appendix

A Data appendix

In this section we describe where we sourced, how we cleaned, and how we constructed the data for this paper. All of the data used in this project is publicly available.

A.1 Protein Data Bank (PDB)

The Protein Data Bank (PDB) was established in 1971. Today, the archive is managed by the Worldwide Protein Data Bank (wwPDB), an international consortium that maintains a single, globally standardized archive of experimentally determined macromolecular structures (Berman et al., 2007). The wwPDB includes regional PDB partners such as the Research Collaboratory of Structural Bioinformatics Protein Data Bank (RCSB PDB), the Protein Data Bank in Europe (PDBe), and the Protein Data Bank in Japan (PDBj), and collaborates closely with related structural biology archives including the Biological Magnetic Resonance Data Bank (BMRB) and the Electron Microscopy Data Bank (EMDB). New structure depositions are submitted through the wwPDB OneDep system and are processed by wwPDB deposition centers according to shared standards, with responsibilities divided geographically (wwPDB consortium, 2019). Additional details about the PDB data can be found on RCSB PDB website.¹ All structures deposited in the PDB are publicly released after a holding period, at which point full metadata and coordinates become accessible. We access the data via the RCSB PDB RESTful API. Documentation can be found at <https://data.rcsb.org/#data-api>.

A.1.1 Data structure

An important point that we abstract away from in the paper is that some protein *structures* are made up of multiple *entities*. A single protein chain folds up into an entity, but in some cases multiple entities bind together to form a multi-entity structure. This complicates some of the data construction in ways we will highlight when relevant.

A single protein structure represents a fairly consistent unit of scientific effort, and so we want to perform our PDB analyses at the structure level. At several points, we need to aggregate some entity-level measures up to the structure level. However, AlphaFold and

¹ <https://pdb101.rcsb.org/learn/guide-to-understanding-pdb-data/introduction>

UniProt/SwissProt are indexed at the entity level. Thus, when building our spillover panel, we structure our analysis around UniProt IDs, which are at the entity level.

A.1.2 Data access and download

We collected PDB data via two separate API endpoints covering all indexed records. Structure-level metadata (also called “entry-level” metadata) including experimental method, deposition and release dates, and linked publication identifiers were downloaded in April 2026 from https://data.rcsb.org/rest/v1/core/entry/{pdb_id}.

Entity-level metadata, including the amino acid sequences and UniProt identifiers were downloaded in April 2026 from https://data.rcsb.org/rest/v1/core/polymer_entity/{pdb_id}/{entity_id}.

A.1.3 Key variables extracted

At the structure level, we extract: structure title, deposit date, release date, experimental data collection date, experimental method, structure solution method for x-ray structures, the source of the starting model used to initialize refinement (template structure), PubMed identifier for linked paper, journal name, and group deposit identifiers.

At the entity level, we extract: UniProt identifier, RCSB cluster similarity measures, standardized amino acid sequence, and sequence length.

A.1.4 Sample construction

The analysis sample is constructed from the merged structure and entity datasets, applying four sequential restrictions:

First, we drop redundant structures. Structures explicitly flagged by depositors as group deposits are thinned to one per group (retaining the first by PDB identifier). We then identify implicit group deposits: any structure whose entities share the same deposit date, same complete author list, and same RCSB 100%-identity cluster assignment (this is a measure of the similarity of the amino acid sequence, described in more detail below). Since the similarity clusters are at the entity level, we compute a structure-level redundancy share and drop structures where more than 50% of entities are flagged as redundant as long as the date and author conditions are met. The motivation for dropping all but one of these group deposits is that we are trying to measure *effort* expended on structure determination. These group deposits lead to hundreds or thousands of near-identical structures, and the effort required is closer to that of a single structure than 100x or 1000x the effort of a single structure.

Second, we drop Covid-19-related structures, identified by keyword matching on the structure title (“covid”, “cov-2”, “sars” combined with “cov”, “coronavirus”).

Third, we restrict to protein structures—keeping only entries classified as “Protein (only)”, “Protein/NA”, or “Protein/Oligosaccharide.”

Fourth, we restrict to structures deposited between January 1, 2017 and March 31, 2025, dropping the most recent twelve months of data to account for the release lag between deposition and public availability. After these four restrictions, the analysis sample contains 71,133 structures.

A.2 Sequence homology and MMseqs2

To classify PDB structures by whether their protein sequences had a solved experimental homolog at the time of deposit, we compute pairwise sequence similarity scores using MMseqs2, a widely used tool for fast protein sequence search and clustering (Steinegger and Söding, 2017). More information, as well as download links for this software, can be found at <https://github.com/soedinglab/MMseqs2>.

A.2.1 Input sequences

Sequences are drawn from the PDB entity-level data (entitySequenceStd). Before running MMseqs2, we apply three restrictions: we retain only entities classified as proteins, drop entities shorter than 10 residues, and drop entities whose sequences consist entirely of the ambiguous residue code “X.”

A.2.2 Computing sequence homology on a rolling basis

For each entity in the PDB, we take its amino acid sequence. Each sequence is compared against a dynamic reference pool consisting of the sequences from all entities from structures released prior to that entity’s own deposit month. This rolling definition ensures the reference pool is contemporaneous with the time of deposit. We use MMseqs2 to perform these comparisons, and it returns the single best match for each query, along with percent of the sequence that is identical.

A.2.3 Aggregation to the structure level

Entity-level similarity scores are aggregated to the structure level by averaging over all entities within a structure, weighting by the number of amino acids in the entity. A structure is classified as having a homolog if the weighted mean identity meets or exceeds 30%, the

conventional threshold in structural biology for being able to use a structure as a template in molecular replacement.

A.3 AlphaFold Protein Structure Database

The AlphaFold Protein Structure Database (AlphaFold DB), maintained by DeepMind and the European Molecular Biology Laboratory’s European Bioinformatics Institute (EMBL-EBI), provides computational three-dimensional structure predictions for hundreds of millions of proteins (Varadi, Anyango, et al., 2022). Each prediction is accompanied by per-residue confidence scores (pLDDT) ranging from 0 to 100. Data can be linked back to the PDB using UniProt identifiers. Note that unlike the PDB, these data are at the entity level.

A.3.1 Data access and download

Metadata on all 200 million plus predicted structures are hosted on Google BigQuery. Because of the size of the dataset, we only download the observations we need. To do this, we query AlphaFold DB with the list of all UniProt IDs linked to our PDB entities to create a dataset that links to our PDB data. More information is available at <https://github.com/google-deepmind/alphafold/blob/main/afdb/README.md>. The BigQuery data can be accessed directly at <gs://public-datasets-deepmind-alphafold-v4>. We queried the data in May 2026.

In addition, there is an option to direct download the subset of AlphaFold DB that links to SwissProt (discussed below). This download is available at <https://alphafold.ebi.ac.uk/download>. We downloaded it in March 2026.

A.3.2 Key variables extracted

The primary variable we use from AlphaFold DB is average predicted local distance difference test (pLDDT). pLDDT is a per amino acid measure of local confidence. It is scaled from 0 to 100, with higher scores indicating higher confidence and usually a more accurate prediction. Average pLDDT averages over all amino acids in the structure.

When merging to the PDB at the structure level, in cases of multi-entity structures, we have a many-to-one merge. When this happens, we average the confidence score, weighting by the number of amino acids in each entity. This preserves pLDDT as the average confidence across each amino acid.

A.4 UniProt and SwissProt

UniProt is the primary repository for protein sequence and functional annotation. It contains information on over 200 million known proteins (The UniProt Consortium, 2025). It consists of SwissProt, which contains manually curated entries reviewed by expert biocurators, and TrEMBL, which contains computationally annotated entries (Bairoch and Apweiler, 2000). Our analysis uses exclusively the SwissProt database.

A.4.1 Data access and download

We downloaded the full SwissProt protein list in February 2024 from the UniProtKB search interface, filtering to reviewed entries: https://www.uniprot.org/uniprotkb?query=*&facets=reviewed%3Atrue. This yields 570,829 UniProt accession numbers, which serves as the universe of proteins for the spillover panel. The set of proteins indexed by SwissProt is mostly fixed over time.

UniProt curators link each protein entry to papers providing evidence for its annotations. We downloaded all publication records associated with SwissProt entries in June 2026 via the UniProt REST API: https://rest.uniprot.org/uniprotkb/{uniprot_id}/publications. This produces a protein-paper linked dataset.

A.4.2 Citation types and category filtering

The API returns citations with a citation type field. We retain only citations of type “UniProt indexed literatures”—papers directly linked to the protein entry by curators or through computationally assisted curation. We exclude large-scale background references that are associated with the database header rather than specific proteins.

Each citation carries one or more category labels indicating the aspect of protein biology it informs: Structure, Function, Sequences, Expression, Subcellular Location, Interaction, Phenotypes & Variants, PTM/Processing, Disease & Variants, Names, and Family & Domains. We parse these and construct binary indicators. Our primary spillover outcome excludes papers tagged with the Structure category, restricting to non-structure papers to ensure the outcome captures downstream scientific engagement rather than structure-determination activity itself.

A.4.3 Papers linking to multiple proteins

A single paper can be linked to multiple SwissProt proteins. To avoid double-counting when aggregating to the protein-year level in our panel, we assign each paper a fractional weight equal to one over the number of SwissProt proteins to which it is linked. For robustness and

for count-based specifications, we also construct raw counts for papers linked to at most 1, 5, 10, or 15 proteins. In these cases, papers will be counted more than once if they link to multiple proteins.

A.5 OpenAlex

OpenAlex is a free, open-source index of scientific publications, maintained by the nonprofit OurResearch. It catalogues papers, authors, institutions, and citation relationships, drawing primarily on data from Crossref, PubMed, and the now-defunct Microsoft Academic Graph (Priem et al., 2022). We use an OpenAlex snapshot downloaded by us in April 2025 and hosted on Google BigQuery.

A.5.1 Data access and download

We begin by taking the full list of SwissProt-indexed papers, and query this list against OpenAlex, using the PubMed IDs provided by SwissProt. We use this to merge on the OpenAlex ID. Call this “crosswalk 1.” We repeat this process with the three key AlphaFold papers that users are encouraged to cite: Jumper et al. (2021), Varadi, Anyango, et al. (2022), and Varadi, Bertoni, et al. (2024). Call this “crosswalk 2.”

Then, we query OpenAlex using OpenAlex IDs, asking to find cases where papers on crosswalk 1 cite papers on crosswalk 2. We can merge this resulting list back to our SwissProt-indexed papers to understand which of the SwissProt papers are citing AlphaFold.

A.6 UniRef and UniClust sequence clusters

To cluster SwissProt proteins based on amino acid sequence, we use sequence-based clustering from the UniRef and UniClust databases. UniRef is part of UniProt, and it provides clustered sets of sequences based on the overlap of the protein’s amino acid sequence using MMseqs2. For example, the UniRef90 cluster data is built by clustering amino acid sequences with at least 11 amino acids that have at least 90% sequence identity and 80% overlap with the longest amino acid sequence in the cluster. We could perform this clustering ourselves using MMseqs2 as described above. However, given the very large size of this database, we prefer to use these pre-constructed cluster measures.

A.6.1 Data access and download

We download the 100%, 90%, and 50% clusters for all structures in SwissProt. These files are sourced from the UniProt ID Mapping tool, the documentation of which can be found

here: https://www.uniprot.org/help/id_mapping. We downloaded these data in April 2024.

UniRef does not host data on 30% clusters, so we sourced this from a UniClust snapshot. Our snapshot of clusters of all SwissProt structures was downloaded in March of 2024. However, the snapshot itself was from 2018. However, because the selection of SwissProt proteins has been very stable over time, we are missing 30% clusters for less than 1% of SwissProt proteins. The most recently-available snapshots are available here: <https://uniclust.mmseqs.com/>.

A.7 SureChEMBL

SureChEMBL is a large, publicly available database of patent documents that have been automatically annotated with chemical and biomedical entities, maintained by EMBL-EBI (Papadatos et al., 2016). An automated text- and image-mining pipeline processes the full text, images, and attachments of patent documents from the major patent authorities and extracts the compounds and—following the SureChEMBL 2.0 update—the genes, proteins, and diseases mentioned in each document.² Protein and gene mentions are recognized using a dictionary-based approach and, where possible, resolved to standard identifiers such as UniProt accessions. Because each annotation links a patent document to a specific protein target, SureChEMBL allows us to measure patenting activity associated with a given protein, which we interpret as a proxy for downstream commercial and applied interest in that protein. Two features of the data are worth noting at the outset. First, the biomedical (non-chemical) annotations are available only for patents that already contain chemical annotations, so our patent sample is implicitly restricted to chemistry-related documents.

A.7.1 Data access and download

SureChEMBL distributes its data as bulk Parquet files through the EMBL-EBI FTP site, which are refreshed on a rolling basis. We downloaded the bulk data in May 2026. Given the size of the data, we hosted the files on the Haas School of Business high-performance computing cluster and performed our extraction using DuckDB. Documentation of the data and its schema is available at <https://chembl.gitbook.io/surechembl>.

The data are organized into several linked tables, of which we use three. The *patents* table contains one row per patent document, with bibliographic and classification metadata. The *biomedical_entities* table contains one row per distinct annotated entity (for example,

² See <https://www.ebi.ac.uk/about/news/updates-from-data-resources/surechembl-2-0-is-here/> for details on the 2.0 update.

a protein), including its type and resolved identifier. The *biomedical_locations* table is a linking table that records which entities appear in which patent documents.

A.7.2 Key variables extracted

Given the size of the data, we wrote a query that joins these three tables and extracts only the observations and variables relevant to our analysis. The query returns one row per patent-protein pair.

Patents. This table is at the patent-document level. For each patent we extract the patent identifier, patent number, country of the issuing authority, publication date, family identifier, patent classification codes (CPC, IPCR, IPC, and ECLA), assignee, and title.

Biomedical locations. This linking table connects patents to the biomedical entities that appear in them. We use it to join each patent to its annotated entities.

Biomedical entities. This table is at the entity level. Each entity carries a type identifier—proteins and genes are resolved to UniProt and HGNC, while diseases are resolved to other controlled vocabularies—together with the text as captured in the patent, a corrected form, and a resolved identifier. We restrict to protein and gene entities and keep the resolved identifier, dropping HGNC (gene-level) identifiers so that we retain UniProt (protein) accessions.

The query links 5,488,183 unique patent documents to 102,398 unique UniProt protein targets, creating a total of 19,585,190 unique pairs.

A.7.3 Data cleaning

From the extracted data we keep, for each patent-protein link, the patent identifier, UniProt identifier, publication date, family identifier, and patent classification codes. We date each patent by its publication year.

We classify patents by technology field using the WIPO IPC-Technology Concordance, which maps patent classification codes to 35 technology fields (Schmoch, 2008). We define a patent as *medical* if any of its classification codes falls in the Medical technology, Biotechnology, or Pharmaceuticals fields. We apply the concordance to a patent’s IPCR codes, falling back to its CPC codes when IPCR is missing.

Because the same underlying invention is often filed as separate patent documents in multiple jurisdictions, we also construct measures based on patent *families*. The family identifier groups all documents that share the same priority claim. We date each family by the earliest publication year among its member documents—its first appearance—and treat a family as medical if any of its member documents is.

A.7.4 Integration into the spillover panel

For each protein we construct annual counts of patent families, dating each family once at its year of first appearance, so that a single invention filed across multiple jurisdictions is not counted more than once. We compute each count for all patents and for medical patents. We collapse these counts to the protein-year level and merge them into our spillover panel using UniProt IDs. For any UniProt ID-years in the spillover panel that do not merge, we impute zeroes.

A.8 Survey of structural biologists

A.8.1 Survey sample

Our sample is comprised of any researcher listed as a last author on a PDB structure between 2017 and April 2026. We focus on last authors, as these are more likely to be principal investigators (PIs). PIs run their own labs and are largely responsible for choosing the research direction of the lab. This criteria gives us a list of 13,746 unique last authors, identified by their last name and first (or first and middle) initial. The average last author has deposited a median of 2 and a mean of 5 structures in our structure analysis sample.

We then attempt to find email addresses for these last authors. We built an automated pipeline that first looks up the PDB deposits associated with each author. It then attempts to find papers linked to these deposits and extract a corresponding author email from DOI/Crossref, Europe PMC, or DOI landing pages. Rows still missing after this structured pipeline were passed to an AI-assisted web search script. The web search passed author name, PDB list, DOI or PubMed ID, and affiliation to OpenAI API model GPT-5.4-mini with web search. GPT-5.4-mini then attempted to find a public email online using this information. In total, 10,782 email addresses were found, a 78% match rate.

We randomly selected 4,700 email addresses and invitations in four waves between June 29 and July 1, 2026. 4,075 were successfully delivered (87%). Reminders were sent between July 6 and July 8. As of July 13, 2026, we had received 452 responses (11% response rate).

A.8.2 Survey implementation

The survey was administered using Qualtrics. The full survey instrument can be found in Appendix D. The survey was deemed exempt by the UC Berkeley IRB (protocol 2026-05-19702).

A.8.3 Analysis sample

Of our 452 responses, we made a series of restrictions. First, we dropped any responses that did not finish the survey (lose 12 responses). Next, we dropped respondents who provided implausible values for our two free-response questions. For Q1, this was any response of more than 500 structures a year (lose six responses). For Q2, this was any response outside of $[0, 100]$ (lose zero responses). Finally, we dropped any respondents who spent less than 90 seconds completing the survey (lose seven responses). This brings our survey analysis sample to 427.

B Classifying proteins as “easy” or “hard” for AlphaFold

AlphaFold2 predicts some protein structures more accurately than others. Confidence scores are one way to measure this. For every amino acid in the protein, AlphaFold2 assigns a predicted Local Distance Difference Test (pLDDT) ranging from 0 to 100, which represents how confident AlphaFold2 is in that amino acid’s placement. Averaging across all amino acids in a protein provides a protein-level measure known as average pLDDT. These scores are informative: on average, proteins with higher pLDDT are ones whose structures AlphaFold2 predicts more accurately, a relationship validated against held-out experimental structures (Jumper et al., 2021; Tunyasuvunakool et al., 2021).

Confidence scores, however, only capture one dimension of how useful AlphaFold2 predictions are. AlphaFold2 has systematic limitations that pLDDT alone does not reveal. Several are worth highlighting.

1. AlphaFold2 models each protein entity in isolation. Many proteins, however, are assemblages of multiple entities. A single entity prediction can be accurate while missing key information about how these entities fit together (Akdel et al., 2022; Terwilliger et al., 2024).³
2. AlphaFold2 only predicts the structure of a protein in isolation, not a protein bound to any other factors. Yet a protein’s functionally relevant shape—especially the geometry of a binding pocket—often depends on what is bound to it, and the unbound (“apo”) prediction can differ from the bound (“holo”) form in ways that matter for biology or drug design (Akdel et al., 2022).
3. Third, AlphaFold2 is poorly suited to predicting how mutations reshape a structure. Feeding it a mutated sequence often returns nearly the same structure, even for point

³ Later models, such as AlphaFold-Multimer and AlphaFold3 have partially filled this gap.

mutations known to disrupt folding (Buel and Walters, 2022).

4. Finally, AlphaFold2 typically returns a single, static structure. This is usually the conformation best represented in its training data. But proteins are flexible and often switch between multiple functional states: open and closed, active and inactive. AlphaFold2 typically reports just one of these, providing no information about possible “fold switching” (Chakravarty et al., 2024).

With these additional weaknesses in mind, we classify proteins as “easy” or “hard” for AlphaFold2 according to the following rules:

- A protein is **easy** if: Its average pLDDT score is greater than 80 AND it is a protein with single entity AND it does not contain any of the following terms in its structure title: {complex, complexed, assembly, heteromer, oligomer, multimer, dimer, trimer, tetramer, antibody, fab, dna-bound, rna-bound, dna bound, rna bound}. All of these terms are meant to capture the weaknesses discussed above. In total, 28% of the proteins in our analysis sample are classified as easy.
- A protein is **hard** if: Its average pLDDT score is less than 80 AND one of the following is true: it has more than one entity OR it is not purely a protein OR it has more than 1000 amino acids OR it contains one of the terms above in its structure title. In total, 14% of the proteins in our analysis sample are classified as hard.

C Supplemental tables and figures

Appendix Table 1: Top 10 4-digit ICPR codes for UniProt-linked patents

ICPR subclass	Subclass title	Share
A61K	Preparations for medical, dental or toiletry purposes	8.6%
A61P	Specific therapeutic activity of chemical compounds or medicinal preparations	7.7%
C12N	Microorganisms or enzymes; compositions thereof; propagating, preserving, or maintaining microorganisms; mutation or genetic engineering; culture media	4.4%
C07K	Peptides	4.2%
G01N	Investigating or analysing materials by determining their chemical or physical properties	3.3%
C07D	Heterocyclic compounds	3.2%
C12Q	Measuring or testing processes involving enzymes, nucleic acids or microorganisms; compositions or test papers therefore; processes of preparing such compositions; condition-responsive control in microbiological or enzymological processes	1.9%
G06F	Electric digital data processing	1.9%
C08L	Compositions of macromolecular compounds	1.7%
Y10T	Technical subjects covered by former us classification	1.5%

Notes: This table reports the most common 4-digit ICPR codes attached to SureChEMBL patents that link to a UniProt ID. Since each patent can have multiple codes, the share represents the share of all 4-digit codes. Each 4-digit code is only counted once per patent.

Appendix Table 2: Top 10 4-digit ICPR codes for UniProt-linked medical patents

ICPR subclass	Subclass title	Share
A61K	Preparations for medical, dental or toiletry purposes	17.8%
A61P	Specific therapeutic activity of chemical compounds or medicinal preparations	16.5%
C12N	Microorganisms or enzymes; compositions thereof; propagating, preserving, or maintaining microorganisms; mutation or genetic engineering; culture media	9.6%
C07K	Peptides	9.4%
C07D	Heterocyclic compounds	4.7%
G01N	Investigating or analysing materials by determining their chemical or physical properties	4.3%
C12Q	Measuring or testing processes involving enzymes, nucleic acids or microorganisms ; compositions or test papers therefor; processes of preparing such compositions; condition-responsive control in microbiological or enzymological processes	4.1%
A61B	Diagnosis; surgery; identification	2.5%
Y02A	Technologies for adaptation to climate change	2.0%
C12Y	Enzymes	1.9%

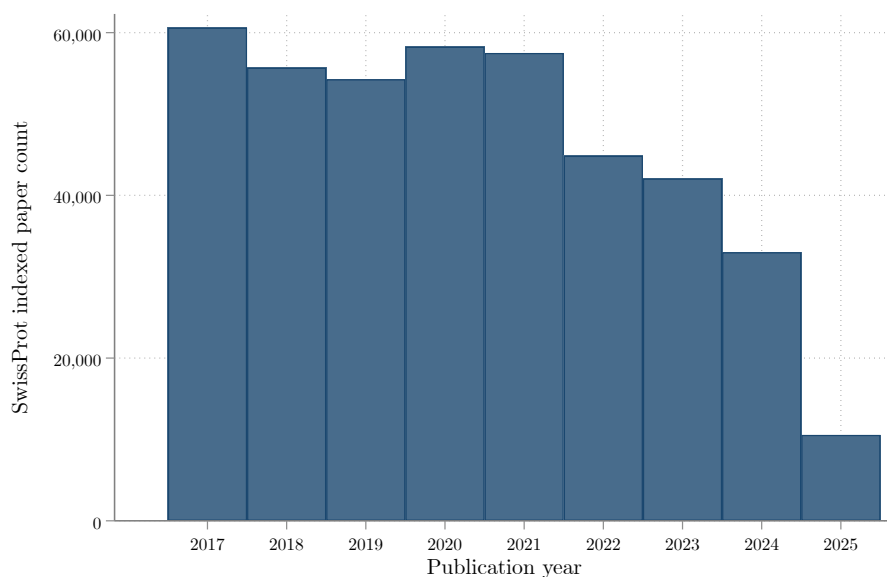
Notes: This table reports the most common 4-digit ICPR codes attached to SureChEMBL medical patents that link to a UniProt ID. Medical patents are defined as patents assigned a ICPR code corresponding to the pharmaceutical, medical technology, or biotechnology fields (Schmoch, 2008). Since each patent can have multiple codes, the share represents the share of all 4-digit codes. Each 4-digit code is only counted once per patent.

Appendix Table 3: Difference-in-differences, robustness by solved definition

Dependent Variable:	(1) Non-structure papers (all paper count)	(2) Non-structure papers (fractional papers)	(3) Non-structure papers (single protein papers)	(4) Patents (all)	(5) Patents (medical)
<i>Panel A. "Unsolved" defined with 30% sequence similarity</i>					
Post	-0.609*** (0.0105)	-0.651*** (0.0131)	-0.649*** (0.0140)	0.173*** (0.0192)	0.022 (0.0145)
Unsolved	-0.991*** (0.0367)	-1.500*** (0.0428)	-1.626*** (0.0458)	-0.394*** (0.1407)	-0.544*** (0.1079)
Post x Unsolved	0.071*** (0.0130)	0.166*** (0.0170)	0.173*** (0.0194)	0.018 (0.0353)	0.016 (0.0208)
Observations	4,901,256	4,901,256	4,901,256	4,901,256	4,901,256
<i>Panel B. "Unsolved" defined with 50% sequence similarity</i>					
Post	-0.646*** (0.0105)	-0.686*** (0.0124)	-0.682*** (0.0134)	0.164*** (0.0212)	0.019 (0.0171)
Unsolved	-1.290*** (0.0365)	-1.768*** (0.0425)	-1.894*** (0.0450)	-0.671*** (0.1282)	-0.824*** (0.1123)
Post x Unsolved	0.130*** (0.0130)	0.216*** (0.0169)	0.222*** (0.0189)	0.029 (0.0334)	0.017 (0.0214)
Observations	4,919,481	4,919,481	4,919,481	4,919,481	4,919,481
<i>Panel C. "Unsolved" defined with 90% sequence similarity</i>					
Post	-0.676*** (0.0117)	-0.712*** (0.0134)	-0.707*** (0.0141)	0.158*** (0.0234)	0.013 (0.0187)
Unsolved	-1.895*** (0.0330)	-2.351*** (0.0391)	-2.488*** (0.0416)	-1.291*** (0.1258)	-1.433*** (0.1197)
Post x Unsolved	0.169*** (0.0136)	0.243*** (0.0167)	0.250*** (0.0182)	0.036 (0.0334)	0.025 (0.0224)
Observations	4,919,481	4,919,481	4,919,481	4,919,481	4,919,481
<i>Panel D. "Unsolved" defined with 100% sequence similarity</i>					
Post	-0.703*** (0.0118)	-0.735*** (0.0131)	-0.728*** (0.0138)	0.145*** (0.0248)	0.012 (0.0195)
Unsolved	-2.248*** (0.0322)	-2.700*** (0.0383)	-2.840*** (0.0408)	-1.668*** (0.1266)	-1.863*** (0.1223)
Post x Unsolved	0.205*** (0.0136)	0.271*** (0.0165)	0.277*** (0.0181)	0.053 (0.0340)	0.026 (0.0229)
Observations	4,919,481	4,919,481	4,919,481	4,919,481	4,919,481

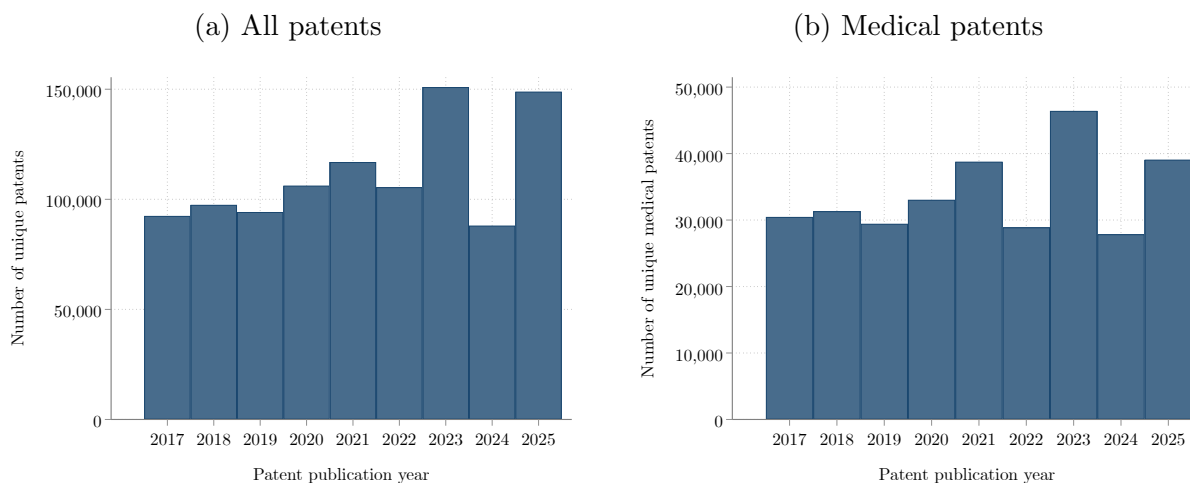
Notes: This table presents Poisson difference-in-differences regression estimates comparing previously solved and unsolved proteins before and after AlphaFold, as in Table 4. Panels A through D use different definitions for “unsolved” based on varying degrees of sequence similarity. The broadest definition is 30% similarity in Panel A. Note the smaller sample size is because 30% clusters were sourced from an older dataset with slightly fewer clusters available. The narrowest definition is 100% similarity in Panel D, which is the preferred definition in the main text. Standard errors are clustered at the cluster level used in each panel. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix Figure 1: SwissProt-indexed publications by publication year



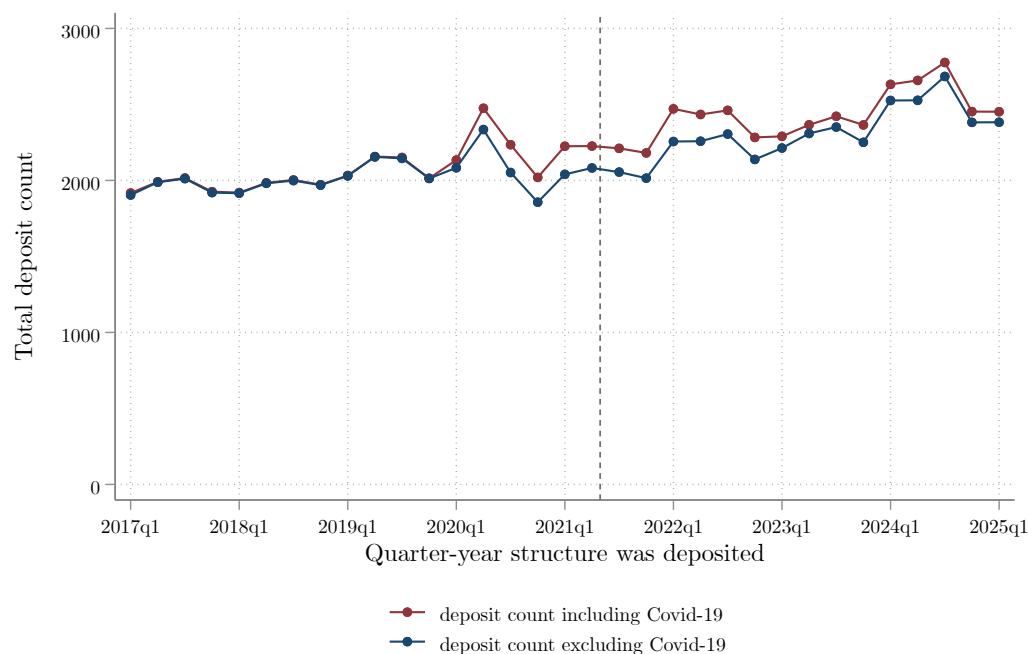
Notes: This figure shows the histogram of publication years for SwissProt-linked papers. $N = 383,625$ papers published between 2017 and 2025.

Appendix Figure 2: SureChEMBL-indexed patents publication year



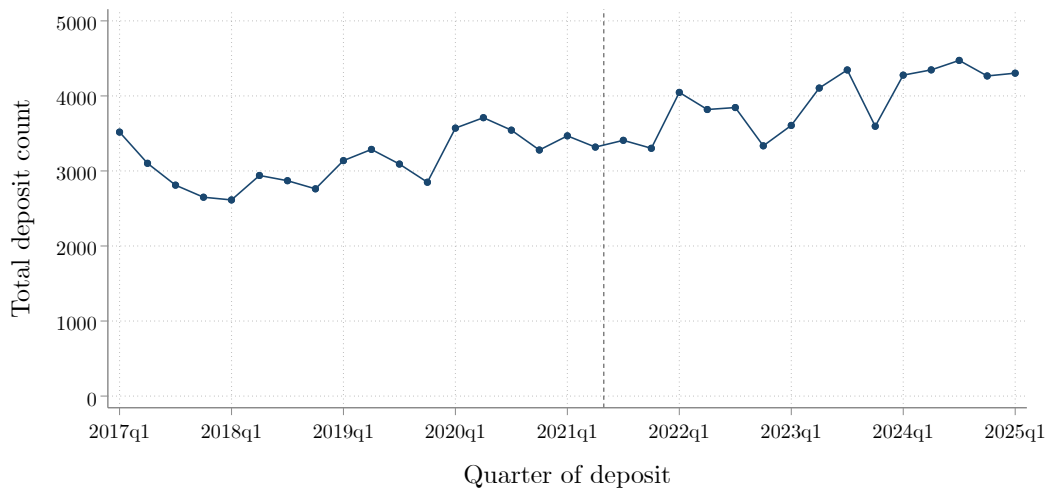
Notes: This figure shows the histogram of publication years for SureChEMBL-linked patents. Patents are collapsed to the family level and assigned the earliest publication date within the family. $N = 2,432,010$ patents and $N = 822,245$ medical patents published between 2017 and 2025.

Appendix Figure 3: Total PDB deposits, including Covid-19 structures



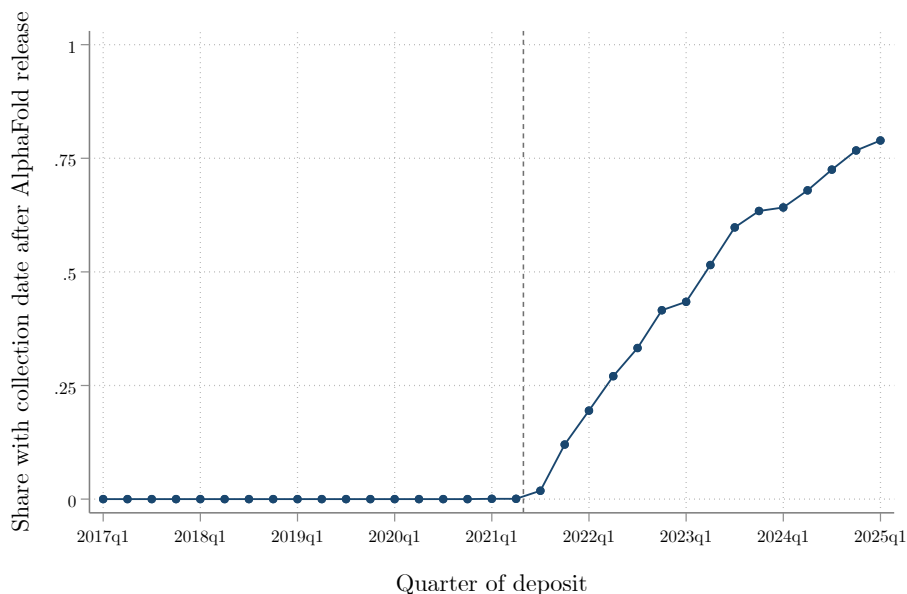
Notes: This figure shows the total counts of PDB structures in our analysis sample over time, adding back in Covid-19 structures. $N = 73,853$ structures.

Appendix Figure 4: Total PDB deposits, no restrictions



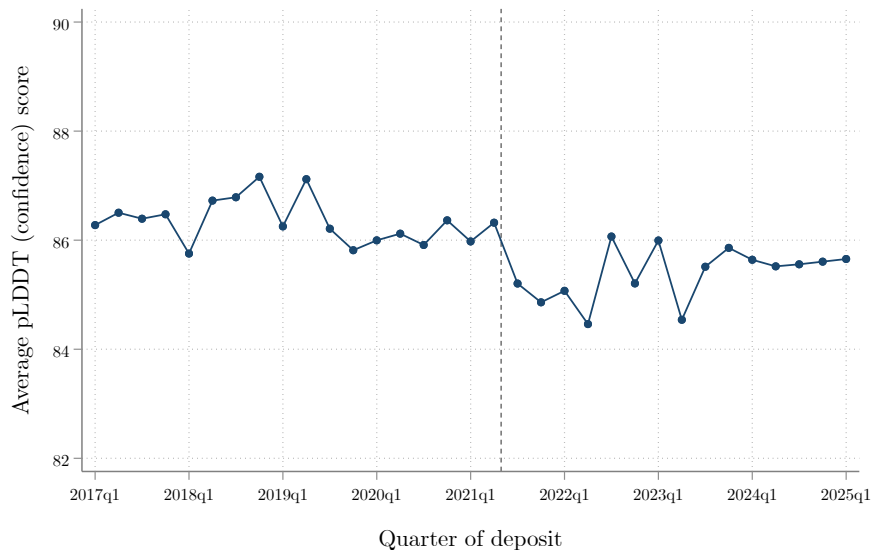
Notes: This figure shows the total counts of PDB structures in our analysis sample over time, making zero sample restrictions. $N = 115,600$ structures.

Appendix Figure 5: Share of structures collecting data after AlphaFold release



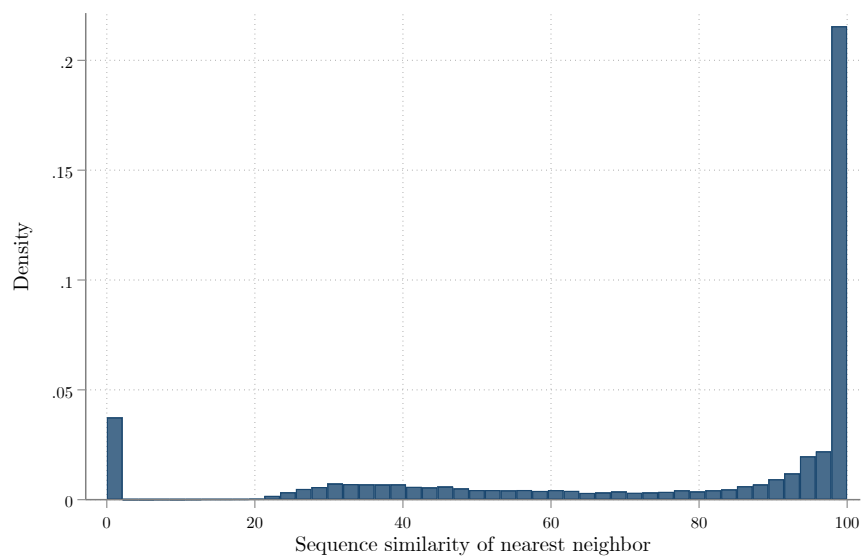
Notes: This figure shows the share of PDB structures that collected their experimental data after AlphaFold's release, by quarter of deposit. $N = 51,696$ structures that report a collection date.

Appendix Figure 6: Average AlphaFold confidence scores of experimentally solved structures



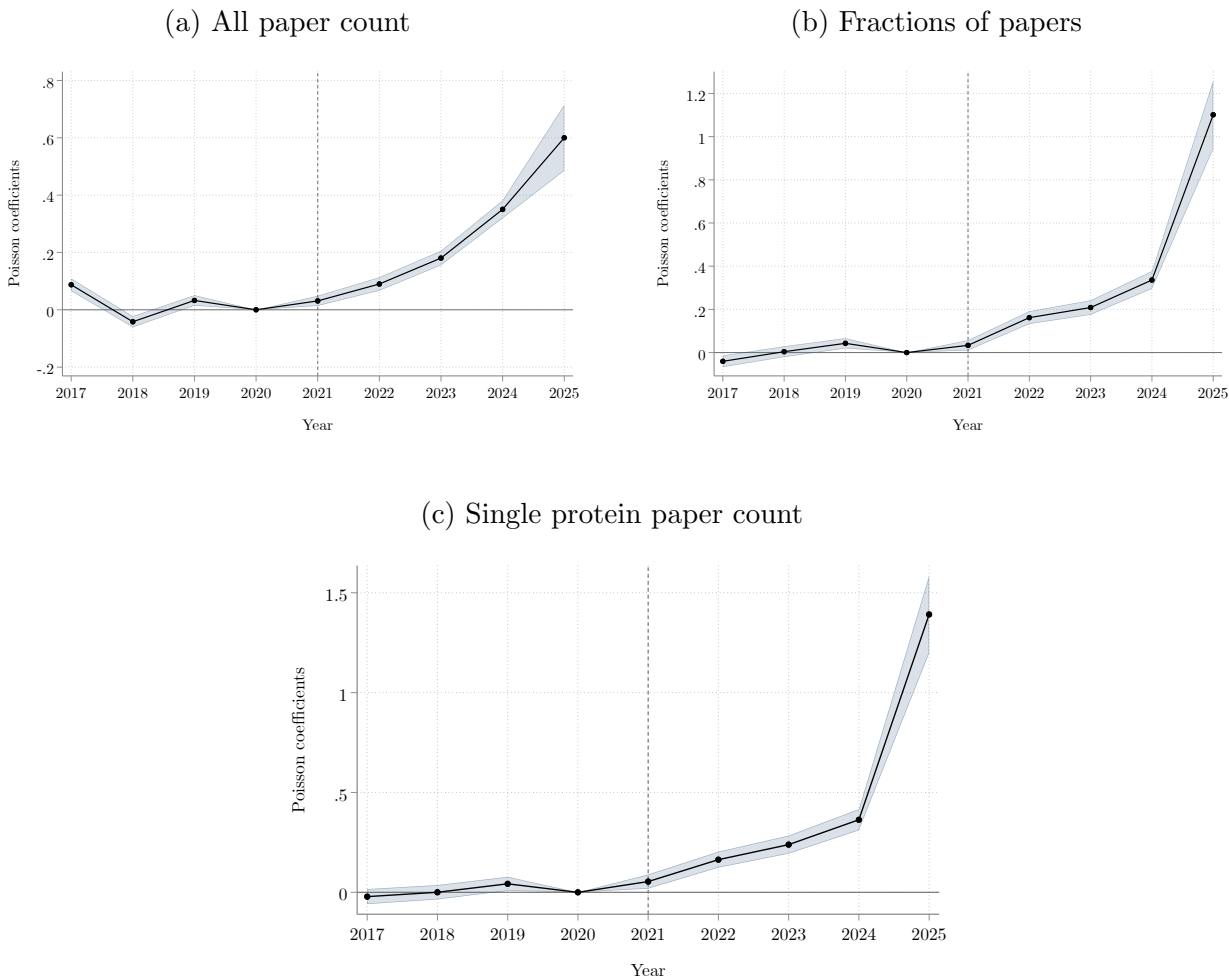
Notes: This figure takes every experimentally solved structure and finds its AlphaFold predicted analog where possible. We then plot the average confidence scores (pLDDT) over time. $N = 59,403$ proteins that have a corresponding AlphaFold confidence score.

Appendix Figure 7: Homology scores of PDB-deposited structures



Notes: This figure shows the distribution of homology scores (sequence similarity of a protein's nearest experimentally-solved neighbor). $N = 51,545$ proteins in the analysis sample solved via x-ray crystallography.

Appendix Figure 8: Non-structure publications: solved vs. unsolved proteins including SARS-CoV-2



Notes: This figure reports the Poisson difference-in-differences coefficients comparing non-structure paper counts for solved vs. unsolved proteins before and after AlphaFold, following Equation 3 in the text. Here we include the 17 proteins in SwissProt that belong to the SARS-CoV-2 virus. Solved structures are defined based on whether the protein had an experimental structure deposited in the PDB prior to AlphaFold's release on July 22, 2021. Panel (a) outcome is counts of all non-structure papers linked to the protein, including those linked to multiple proteins. Panel (b) outcome is fractional papers, defined as one divided by the number of distinct proteins linked to the non-structure paper. Panel (c) outcome is counts of non-structure papers that map uniquely to a single protein. Confidence intervals are shaded in blue and are calculated with standard errors clustered at the protein level. $N = 4,919,634$ SwissProt protein-years, coming from 546,626 SwissProt proteins.

D Survey instrument

The full text of the Qualtrics survey instrument can be found below:

In these two questions, we will ask you about three different time periods. Please provide an answer for each time period.

How many experimental protein structures does your lab solve per year?

Before AlphaFold2 (2017 to 2020) _____

After AlphaFold2 (2021 to present) _____

Five years from now (2031, expected) _____

What percent of your lab's total research time is devoted to experimental protein structure determination? (0 to 100)

Before AlphaFold2 (2017 to 2020) _____

After AlphaFold2 (2021 to present) _____

Five years from now (2031, expected) _____

In the following three questions, indicate whether you agree or disagree with the following statements:

AlphaFold2 and related advances in protein structure prediction have caused me to shift my research focus toward experimentally solving proteins that are predicted poorly by AI methods.

- ☐ Strongly agree
- ☐ Somewhat agree
- ☐ Neither agree nor disagree
- ☐ Somewhat disagree
- ☐ Strongly disagree

Since the development of AlphaFold2 and related developments in protein structure prediction, it has become harder to publish experimental protein structures in top scientific journals.

- ☐ Strongly agree
- ☐ Somewhat agree
- ☐ Neither agree nor disagree
- ☐ Somewhat disagree
- ☐ Strongly disagree

Since the development of AlphaFold2 and related developments in protein structure prediction, it has become harder to secure grant funding for experimental protein structure determination.

- ☐ Strongly agree
- ☐ Somewhat agree
- ☐ Neither agree nor disagree
- ☐ Somewhat disagree
- ☐ Strongly disagree

Do you trust AlphaFold2 or other protein structure predictions enough to skip experimental validation?

- ☐ Yes, always
- ☐ Yes, most of the time
- ☐ Sometimes
- ☐ No, usually not
- ☐ No, never

Would you recommend that current students continue to invest in learning experimental structural biology techniques such as X-ray crystallography or cryo-EM?

- ☐ Definitely yes
- ☐ Probably yes
- ☐ Might or might not
- ☐ Probably not
- ☐ Definitely not

Have AlphaFold2 and other protein structure prediction tools made your job more or less fulfilling?

- ☐ Much more fulfilling
- ☐ Slightly more fulfilling
- ☐ Neither more nor less fulfilling
- ☐ Slightly less fulfilling
- ☐ Much less fulfilling

Do you have any other comments that would help us better understand how AlphaFold2 and related protein structure prediction tools are impacting experimental structural biology and related work?

References

- Akdel, Mehmet et al. (2022). “A Structural Biology Community Assessment of AlphaFold2 Applications”. *Nature Structural & Molecular Biology* 29.11, pp. 1056–1067.
- Bairoch, Amos and Rolf Apweiler (2000). “The SWISS-PROT protein sequence database and its supplement TrEMBL in 2000”. *Nucleic Acids Research* 28.1, pp. 45–48.
- Berman, Helen et al. (2007). “The worldwide Protein Data Bank (wwPDB): ensuring a single, uniform archive of PDB data”. *Nucleic Acids Research* 35.Database issue, pp. D301–D303.
- Buel, Gwen R and Kylie J Walters (2022). “Can AlphaFold2 predict the impact of missense mutations on structure?” *Nature Structural & Molecular Biology* 29.1, pp. 1–2.
- Chakravarty, Devlina et al. (2024). “AlphaFold predictions of fold-switched conformations are driven by structure memorization”. *Nature Communications* 15.1, p. 7296.
- Jumper, John et al. (2021). “Highly accurate protein structure prediction with AlphaFold”. *Nature* 596.7873, pp. 583–589.
- Papadatos, George et al. (2016). “SureChEMBL: a large-scale, chemically annotated patent document database”. *Nucleic Acids Research* 44.D1, pp. D1220–D1228.
- Priem, Jason, Heather Piwowar, and Richard Orr (2022). “OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts”. *arXiv*. arXiv: [2205.01833](https://arxiv.org/abs/2205.01833) [cs.DL].
- Schmoch, Ulrich (2008). *Concept of a Technology Classification for Country Comparisons*. Final Report to the World Intellectual Property Organisation (WIPO). Prepared by Fraunhofer Institute for Systems and Innovation Research (ISI). Karlsruhe: World Intellectual Property Organization (WIPO).
- Steinegger, Martin and Johannes Söding (2017). “MMseqs2 Enables Sensitive Protein Sequence Searching for the Analysis of Massive Data Sets”. *Nature Biotechnology* 35.11, pp. 1026–1028.
- Terwilliger, Thomas C. et al. (2024). “AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination”. *Nature Methods* 21.1, pp. 110–116.
- The UniProt Consortium (2025). “UniProt: the Universal Protein Knowledgebase in 2025”. *Nucleic Acids Research* 53, pp. D609–D617.
- Tunyasuvunakool, Kathryn et al. (2021). “Highly accurate protein structure prediction for the human proteome”. *Nature* 596.7873, pp. 590–596.
- Varadi, Mihaly, Stephen Anyango, et al. (2022). “AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models”. *Nucleic Acids Research* 50.D1, pp. D439–D444.
- Varadi, Mihaly, Damian Bertoni, et al. (2024). “AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences”. *Nucleic Acids Research* 52.D1, pp. D368–D375.
- wwPDB consortium (2019). “Protein Data Bank: the single global archive for 3D macromolecular structure data”. *Nucleic Acids Research* 47.D1, pp. D520–D528.